

BS-480&BS-490 Installation Guide

Department of Biochemistry Global Technical Support

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1 Before Installation

1.1 Installation Environment

- The system is for indoor use only.
- The bearing platform (or ground) should be level (with gradient less than 1/200).
- The bearing platform (or ground) should be able to support at least 300Kg weight.
- The installation site should be well ventilated.
- The installation site should be free of dust.
- The installation site should not be in direct sun.
- The installation site should be kept away from a heat or draft source.
- The installation site should be free of corrosive gas and flammable gas.
- The bearing platform (or ground) should be free of vibration.
- The installation site should be kept away from large noise and power supply interference.
- Keep the system away from brush-type motors and electrical contact device that is frequently switched on and off.
- Do not use such devices as mobile phones and radio transmitter near the system.
- The system should be installed in a place with altitude height between -400 to 2,000 meters. In places with an altitude height between 2,000m to 4,000m, an external vacuum pump should be employed.
- Operating temperature: 15°C-30°C with fluctuation <2°C/H.
- Relative humidity: 35%-85%, without condensation.
- Provide air conditioning equipment if the room temperature does not meet the requirements.
- Install the equipment near the sewer for convenient discharging of waste.

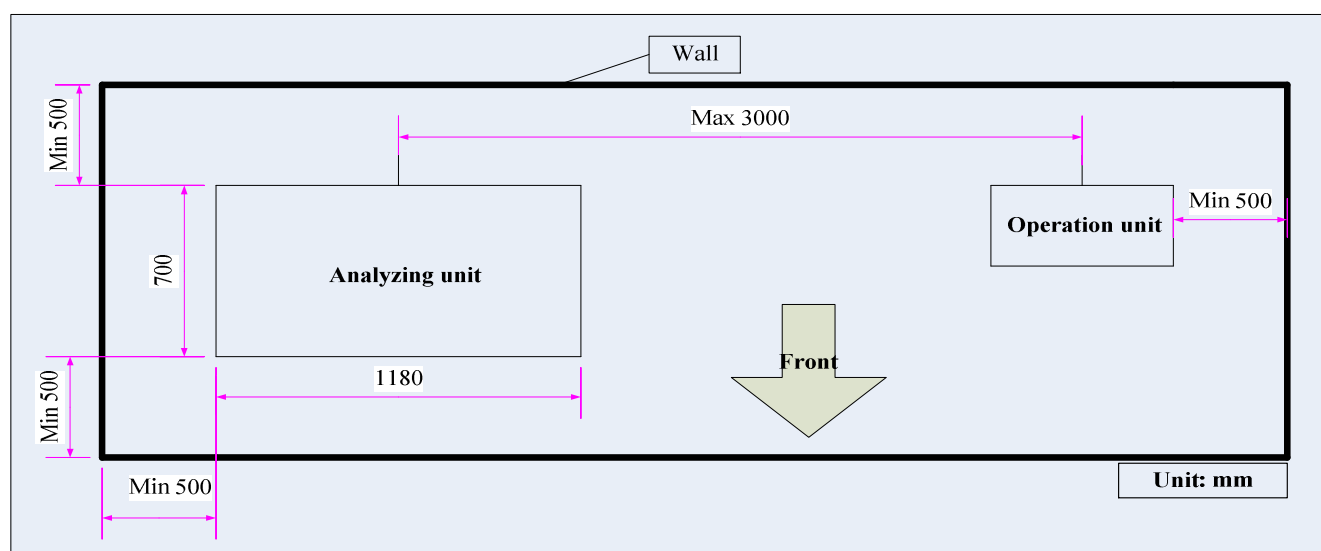
1.2 Dimensions

- Equipment with crate: 1530mm(L) × 935mm(W) × 1390mm(H)
- Equipment without crate: 1180mm(L) × 700mm(W) × 1145mm(H)

1.3 System Configuration

- BS-480/BS-490 chemistry analyzer
- Accessory kit
- Water Supply Unit
- ISE accessory kit (for ISE module only, Optional)
- External air pump (Optional)

1.4 Space and Accessibility Requirements



1.5 Power Supply and Noise

- Power supply: 220V: 220-240V~ 50Hz, or 220/230V~ 60Hz; 110V: 110/115V~ 60Hz. Voltage fluctuation: +/-10%. Line frequency: +/-1Hz. Three-wire power cord with good grounding performance.
- The instrument should be powered by a properly-grounded power supply no less than 3000VA, and the main socket used to connect the instrument should have its load no less than 10A.
- Keep the instrument at least 2.5m away from the main socket.
- If you are going to use a UPS to power the instrument, make sure that the UPS can provide power supply no less than 3000VA/2100W.

**Warning**

Make sure the power socket is grounded correctly and has a proper fuse installed. Improper grounding may lead to electric short or equipment damage.

Put a multimeter on 250V AC range to check if the power socket output voltage meets the specified requirements. When it is well grounded, the voltage between Neutral and Ground should be less than 5V, and the voltage between Live and Ground should be approximately equal to the voltage between Neutral and Ground.

1.6 Water Purifier Specification

- Water quality: meeting requirements of CLSI type II with specific resistance no less than 1 (MΩ.cm@25°C) and with silicate less than 0.1mg/L.
- Water Temperature: 5 - 32°C.
- Output of water: continuous flow no less than 38L/H; peak flow no less than 2L/Min.
- Water supply pressure: 49-392kPa. Mindray water supply unit is standard configuration.
- Water tank: non-pressurized, float-switch-controlled, 38L at least. Water supply unit should provide the DI water to BS-480 from this tank.
- Tube adaptor: Please prepare a proper adaptor to connect the Tank with BS-480 water supply unit. The inside diameter of inlet tube of BS-480 water supply is 4mm. The outside diameter is 6mm.

**BIOHAZARD**

Dispose of the waste liquid according to the local regulations.

**CAUTION**

The supplied water must meet the requirements of CLSI type II; otherwise insufficiently purified water may result in misleading test results.

1.7 Computer Configuration

1.7.1 CPU

At least Intel Dual Core, 8400 @ 3.0GHz

1.7.2 Memory

At least 2GB for each memory

1.7.3 Screen Resolution

The screen resolution should support 1280*1024 pixels and color quality as Highest (32 bit).

1.7.4 Net Port

In case you are going to connect the computer with the LIS or Internet, you should equip the computer with net port (Intel gigabit network adapter).

1.7.5 Serial Port

The computer should provide an RS232 serial port, which is used to connect it with the chemistry analyzer.

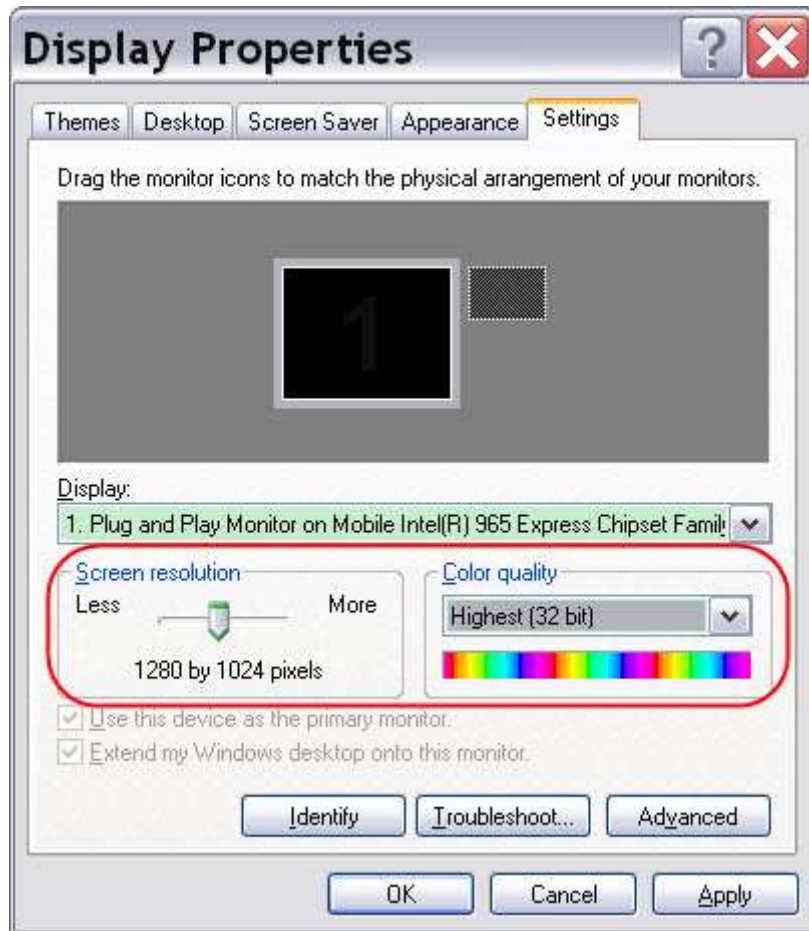
1.7.6 Windows Operating System

The Windows operating system installed on the computer must be **an activated Microsoft Windows 7 Home (32bit)**.

Pre-set the Computer

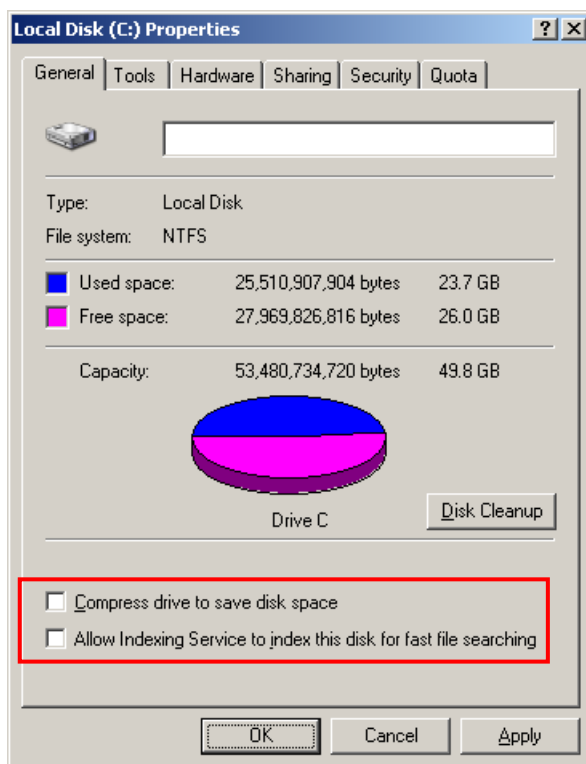
1.7.7 Set the Screen Resolution

Set the screen resolution should support 1280*1024 pixels and color quality as Highest (32 bit).



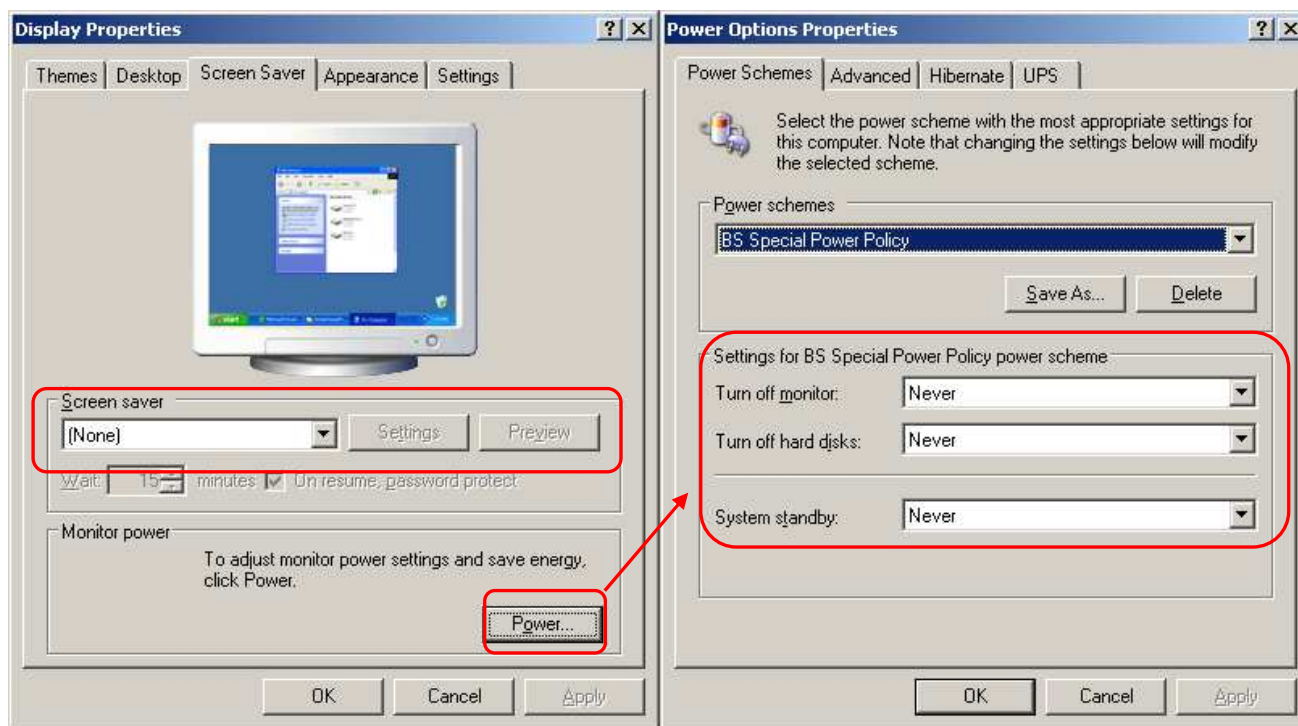
1.7.8 Hard Disk Defragment

Install the Window 7 system in the C drive and the operation software of the instrument in the D drive. Make sure that the C drive is over 30G and D drive is over 100G, and the disk file system is NTFS format. Deselect the two options at the bottom of the disk properties window: **“Compress drive to save disk space”** and **“Allow Indexing Service to index this disk for fast file searching”**.

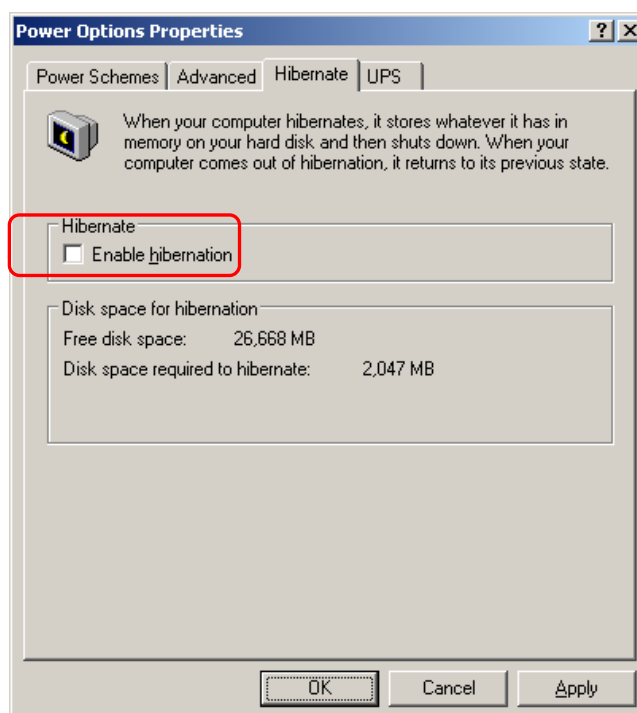


1.7.9 Remove Screen Saver and System Standby

To remove the screen saver and system standby, right click on the desktop, select Properties to display the Display Properties window. Select **None** in the Screen saver pull-down list box, and then select the Power button to go to the Power Options Properties window. Perform the settings as shown in the figures below.

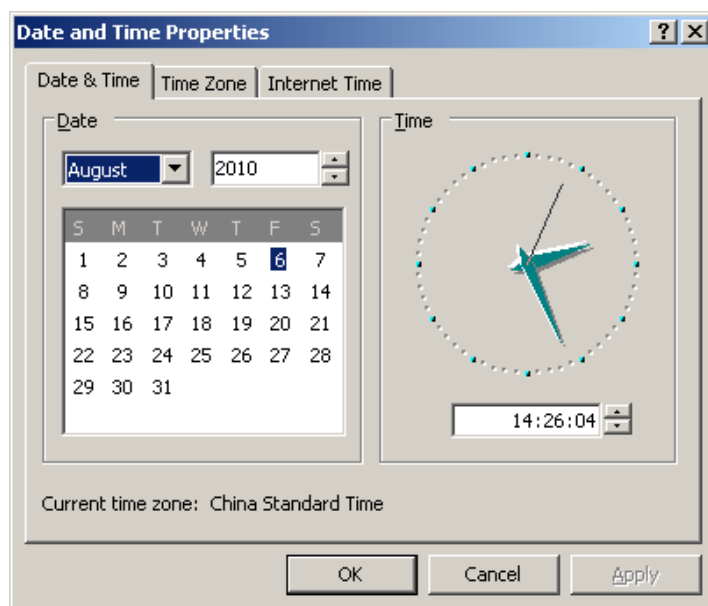


Select the Hibernate page, and then deselect the **Enable hibernation** checkbox.

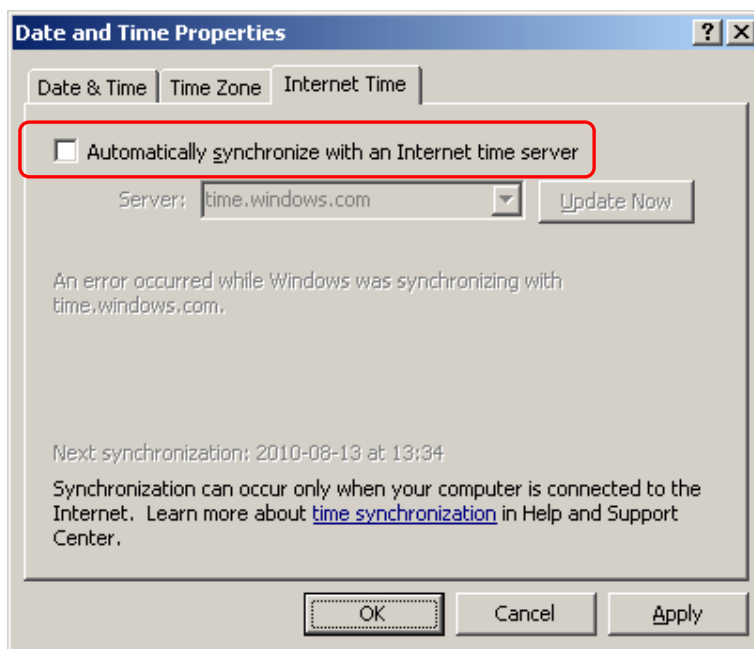


1.7.10 Disable Automatic Synchronization with Internet Time Server

Double click the time icon on the bottom-right corner of the task bar. The Date and Time Properties window is displayed.

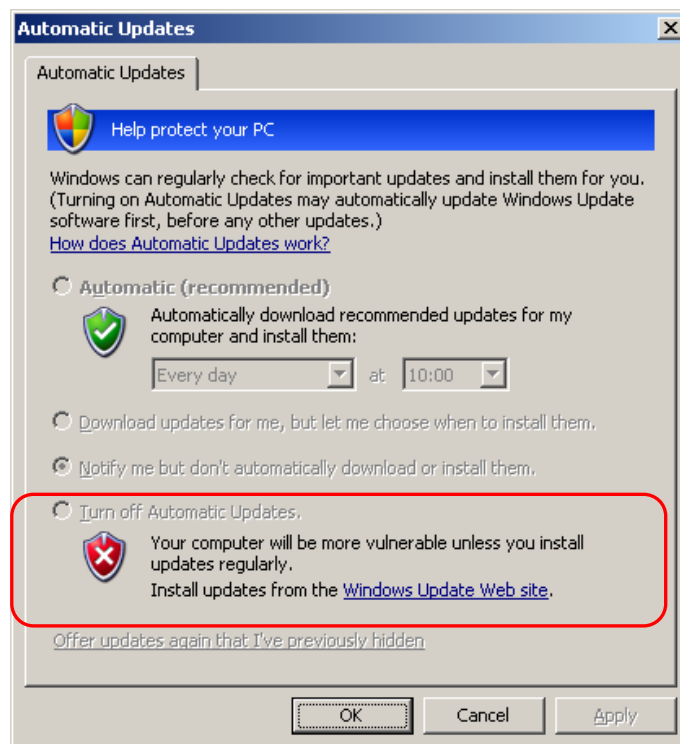


Select the Internet Time tab, and then deselect the checkbox of **automatically synchronize with an Internet time server**.



1.7.11 Turn off Automatic Updates

Turn off the automatic updates on the Automatic Updates window.



1.7.12 Configure Other Application Software

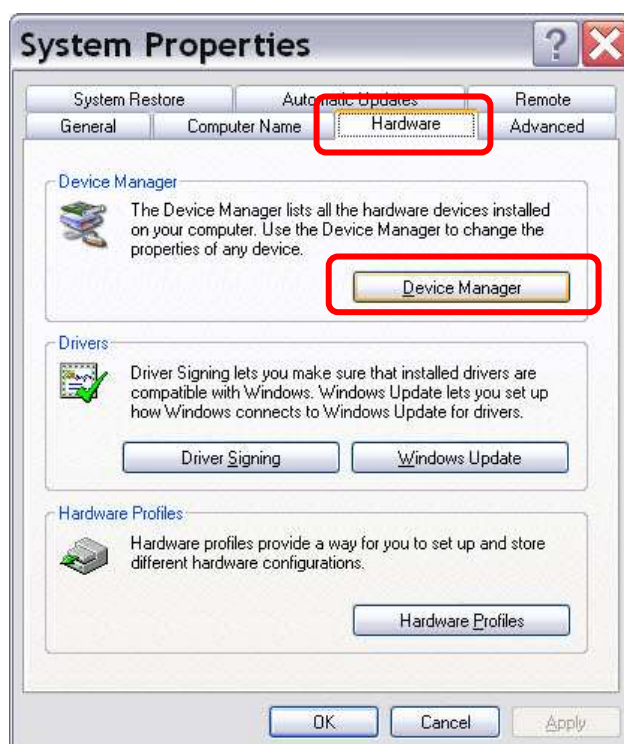
Except for the BS-480 operation software, other software must NOT be installed or reserved on the computer. If an anti-virus software has been installed, remove the automatic scanning and add two process “BS800” and “BSLOG” to its trustable program list after installing the operation software.

1.7.13 Set the COM Port Information

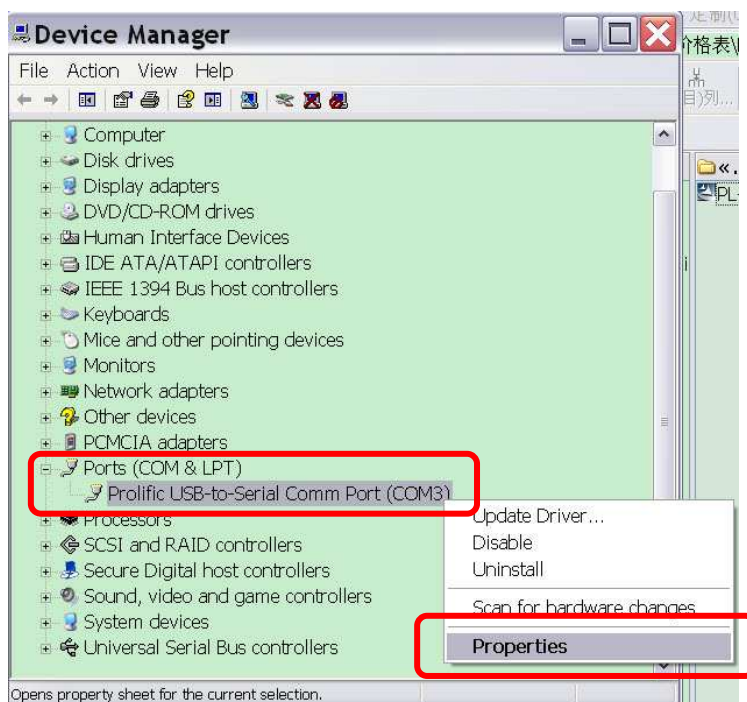
1. Right click “My Computer” icon and then choose “Properties”.



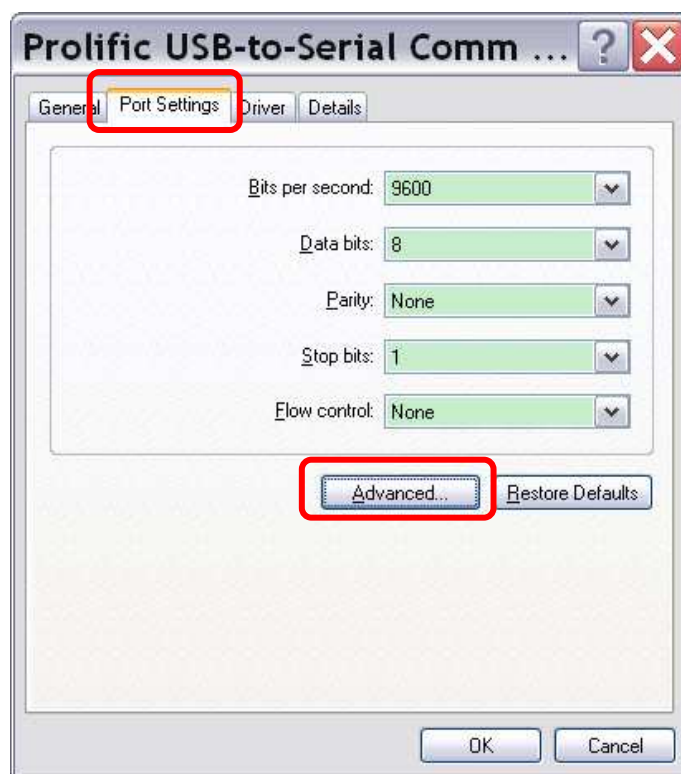
2. Access “Hardware”->”Device Manager”.



3. In “Ports”, find your COM port information, right click it and choose “Properties”.

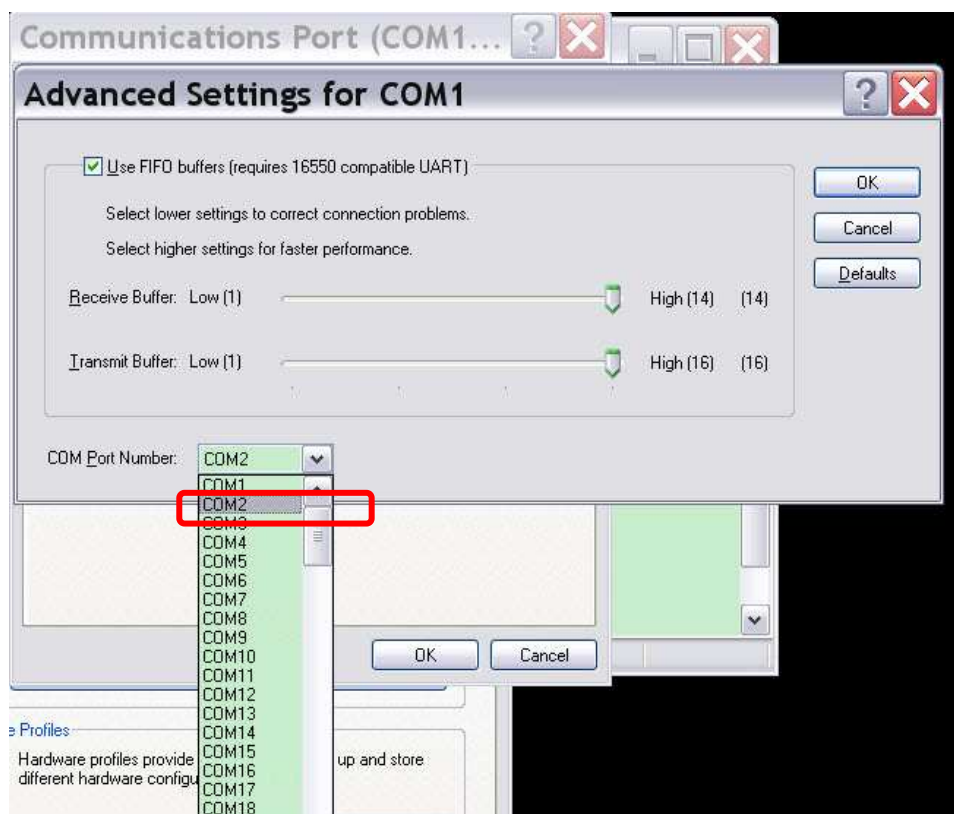
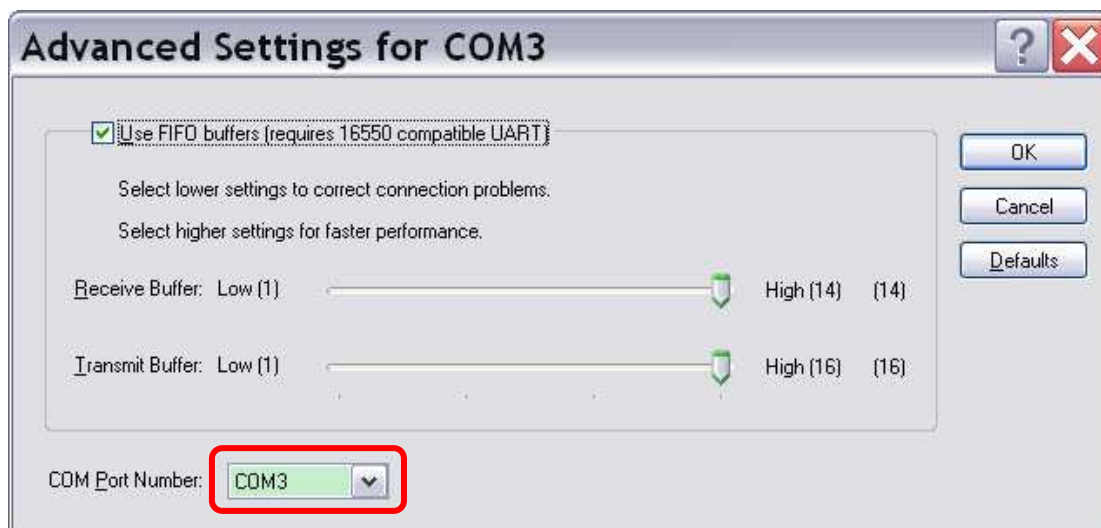


4. Access to “Port Settings”->”Advanced”.



5. If BS-480 with SN **before** XW-34000091/XX-34000025, change the “COM Port Number” to

“COM2”, BS-480 with SN after XW-34000091/XX-34000025, change the “COM Port Number” to “COM1”, then click “OK”.



6. Close “Device Manager” then repeat step 3 and 4 to re-open “Device Manager”. Check if the port is set to “COM2” or “COM1” successfully. If not, repeat step 3~6.

2 Install the Operation Software

2.1 Preparation

1. In case software installation is failed, you have to format the partition C, reinstall Windows system and retry installing BS-480 operation software, which is time-consuming. So it is highly suggested to make a **Windows system ghost file** before you start installing the operation software. You may search online or ask your IT department for more details.
2. For new installation, please use the **software CD in accessory kit** instead of installing the latest operation software which is downloaded from Mindray Technical Document Platform (TDP).

2.2 Installation

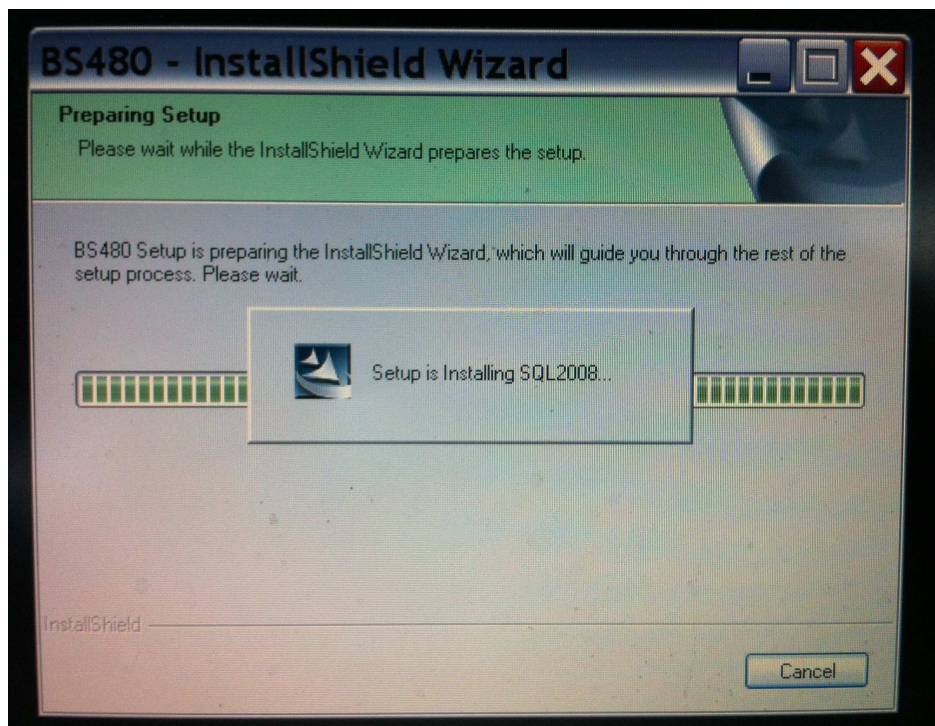
Make sure that there is not any software except the windows system. Copy the installation file from CD to D drive.

Run the setup.exe file in the Setup folder and then follow the steps below.

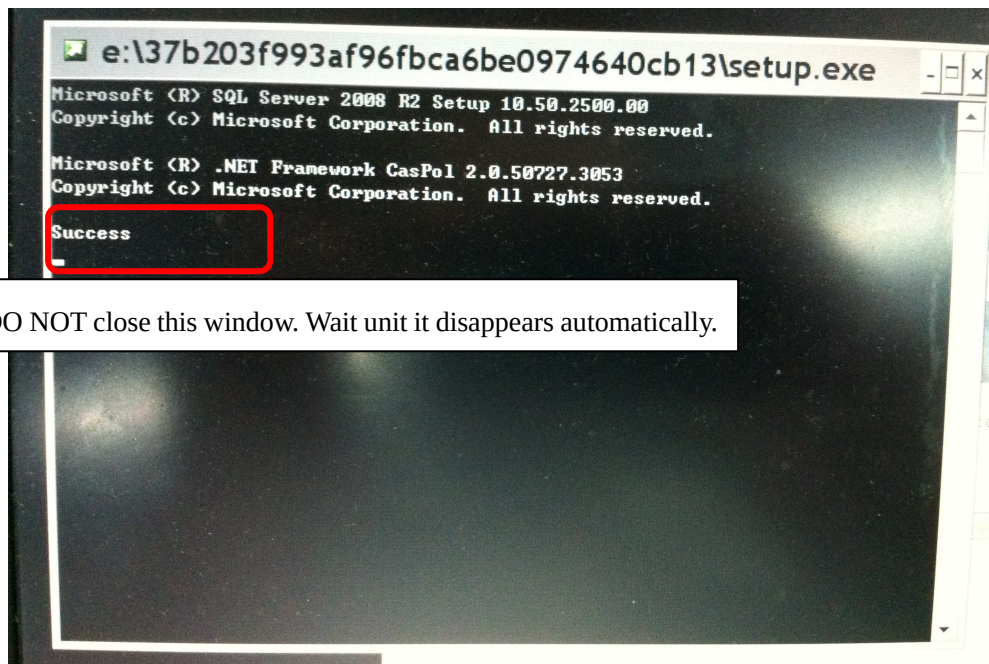
1. Select **Next**.**2.** Then it will install MSI and .Net framework automatically. This procedure will take around 15~20minutes, please be patient. After this step is finished, **PC will reboot automatically.**



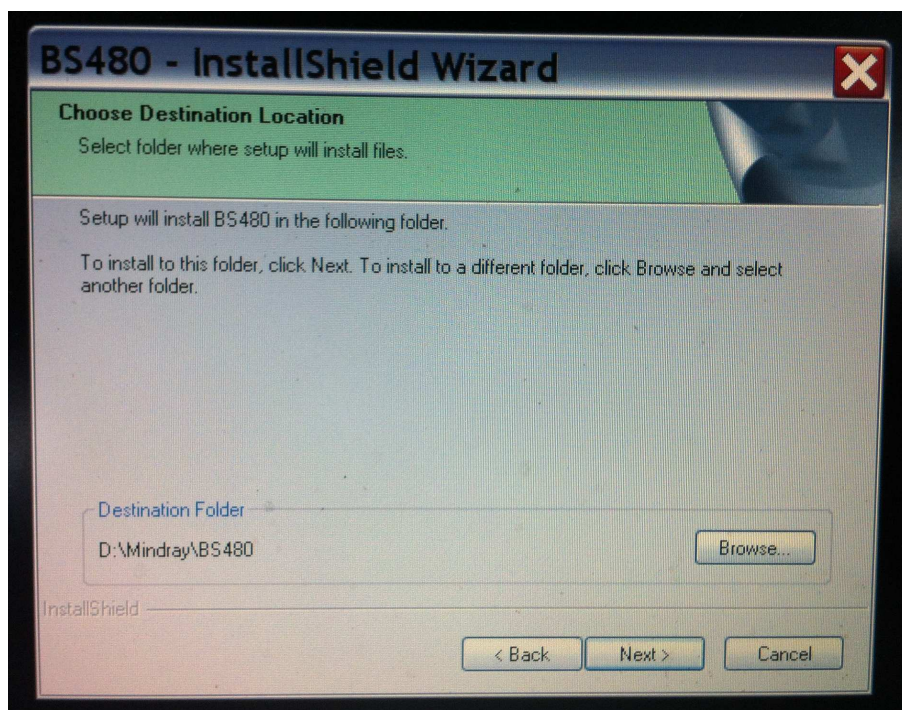
3. After reboot, software will install SQL2008 automatically.



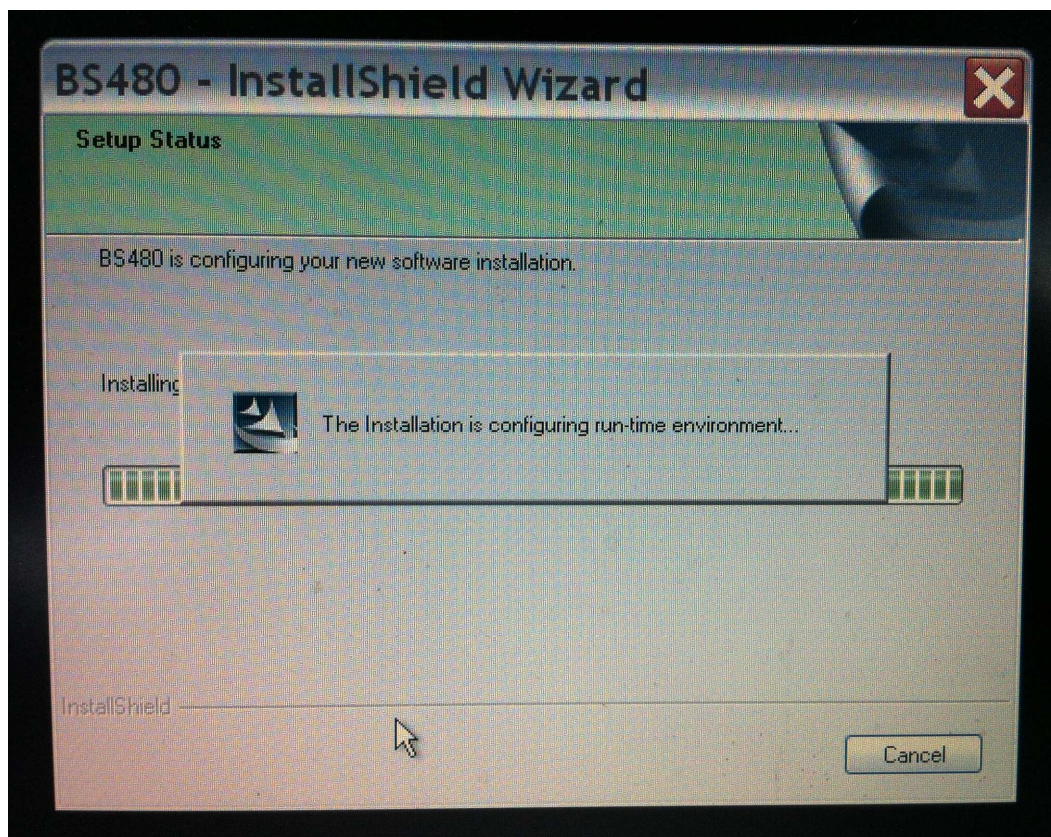
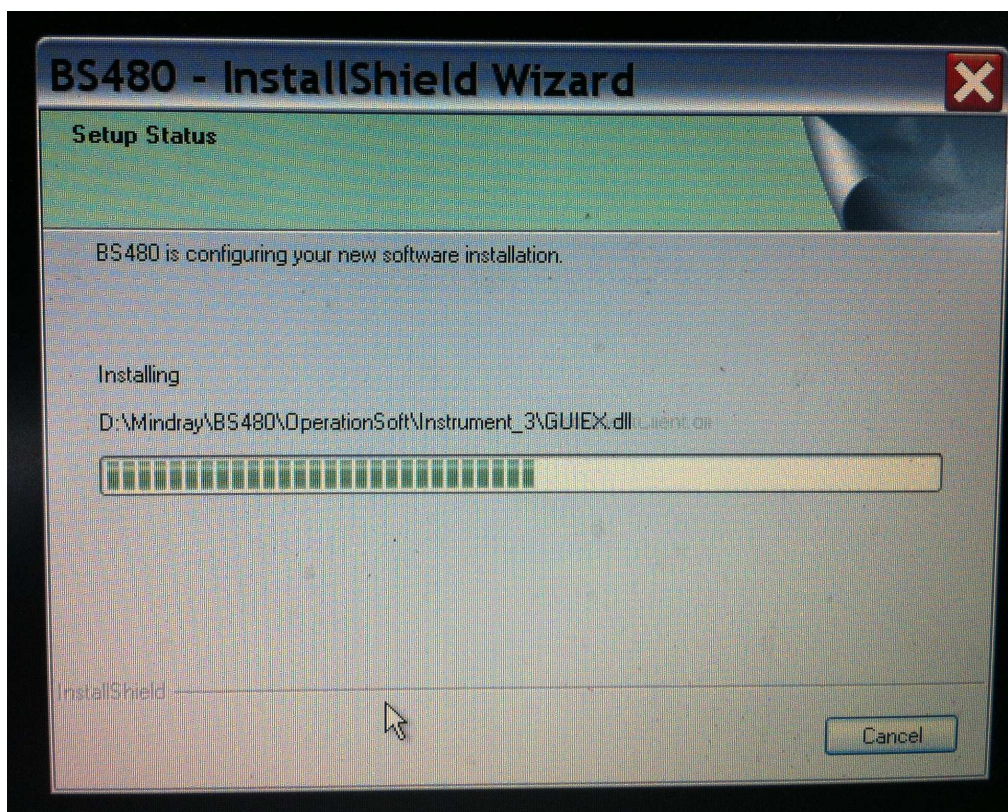
4. You will see a window blow. **DO NOT close this window even you see “Success”.** Just wait around 15 minutes, this window should disappear automatically.

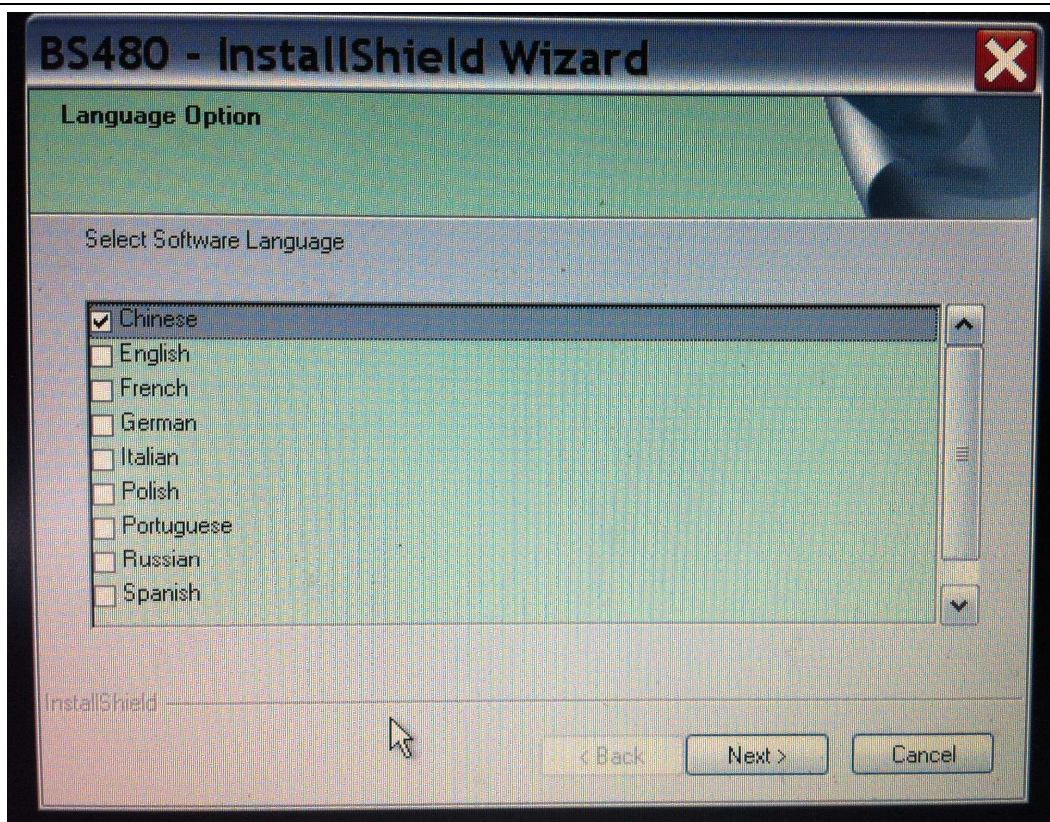


5. When the window disappears, you will see another message to indicate the installation path. DO NOT change the default path. Click **Next** to continue.

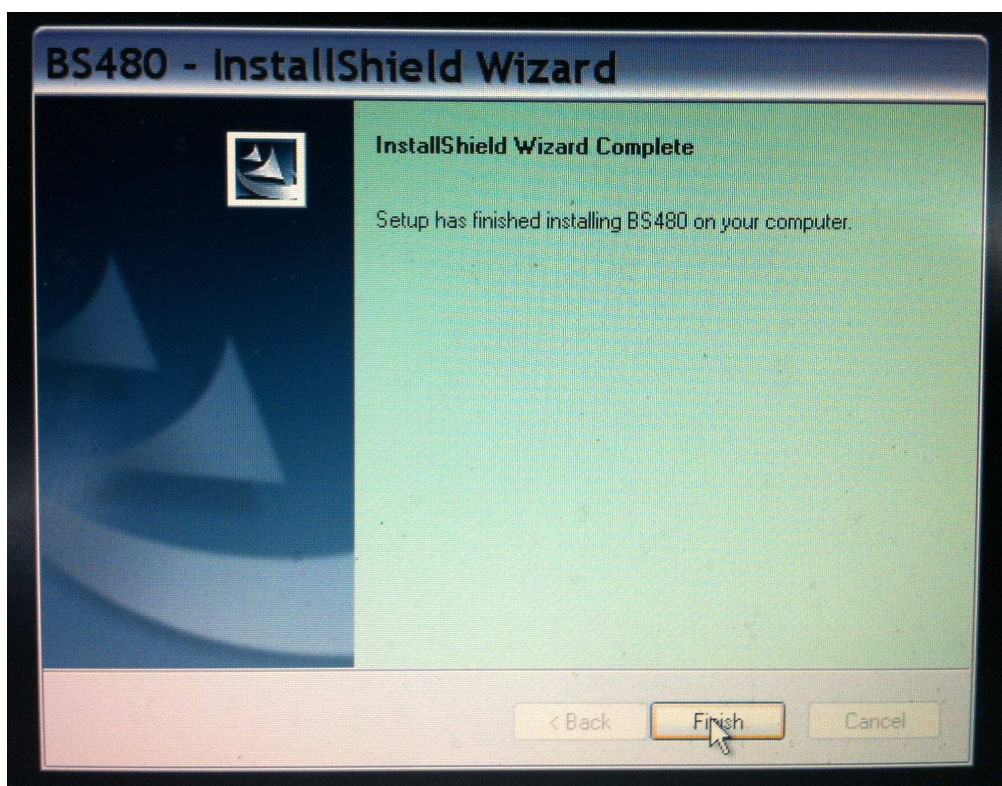


6. The installation continues and you may choose language later to display the software and print test reports.



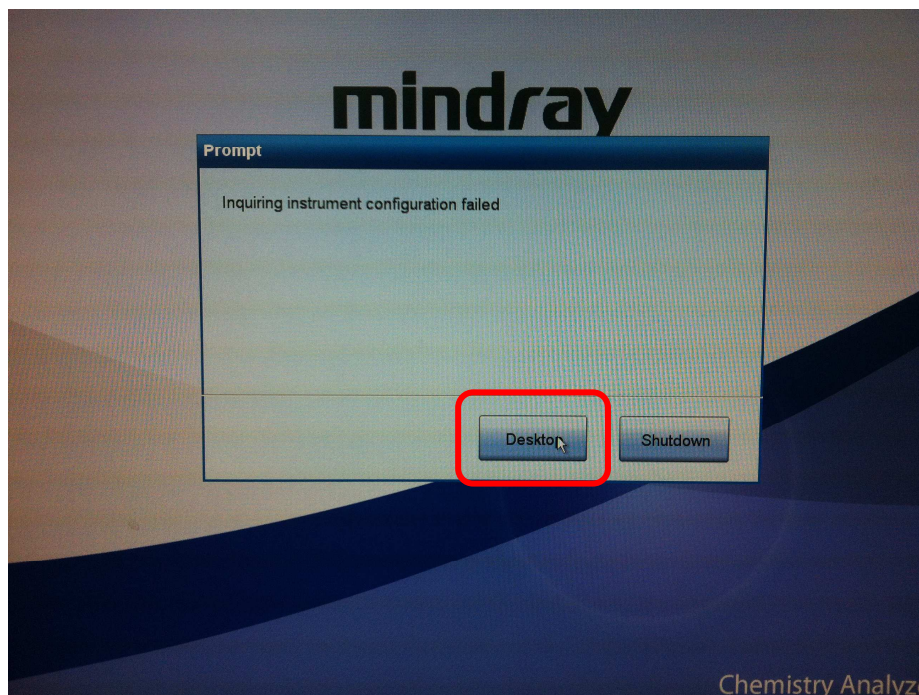


7. Click **Next** to continue and click **Finish** to complete the software installation.

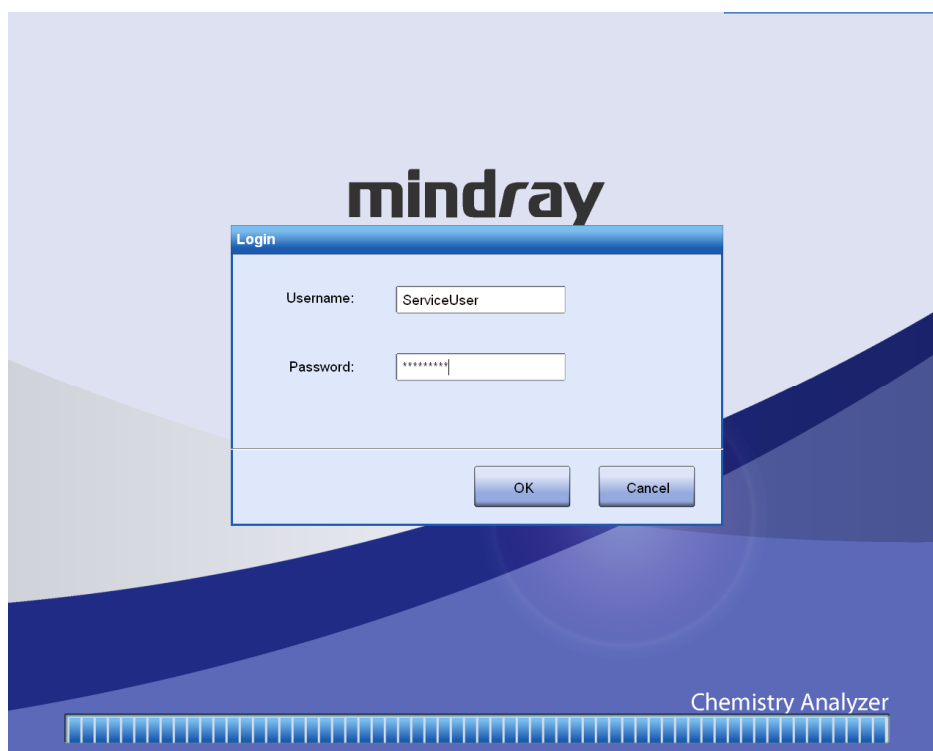


2.3 Software Installation Verification

1. After installation, the operation software will run automatically. Due to disconnecting with the equipment, the software will prompt a message below. It is normal, just click **Desktop**.



2. When the login window appears, enter the username “**ServiceUser**” and password “**#BS8A#SEU**”.
Click **OK** to return to Windows desktop.



-
-
- 3.** Operation software installation has been finished successfully. You may shut down PC now. To run the operation software, just reboot PC.

3 Hardware Installation

3.1 Unpacking Tool

- 1 Big wrench: 30mm for rotating the anchor of BS-480; 30mm can cover BS-480 and BS-800.
- 1 Measure tape: 3M for measuring the space
- 1 Knife for unpacking
- 1 Diagonal cutting pliers for cutting tube(OD 18mm)
- 1 Multimeter
- Common tool: screwdriver, Allen key
- Special Tool: socket spanner(size:14mm) and extender as follows.



3.2 Unpacking

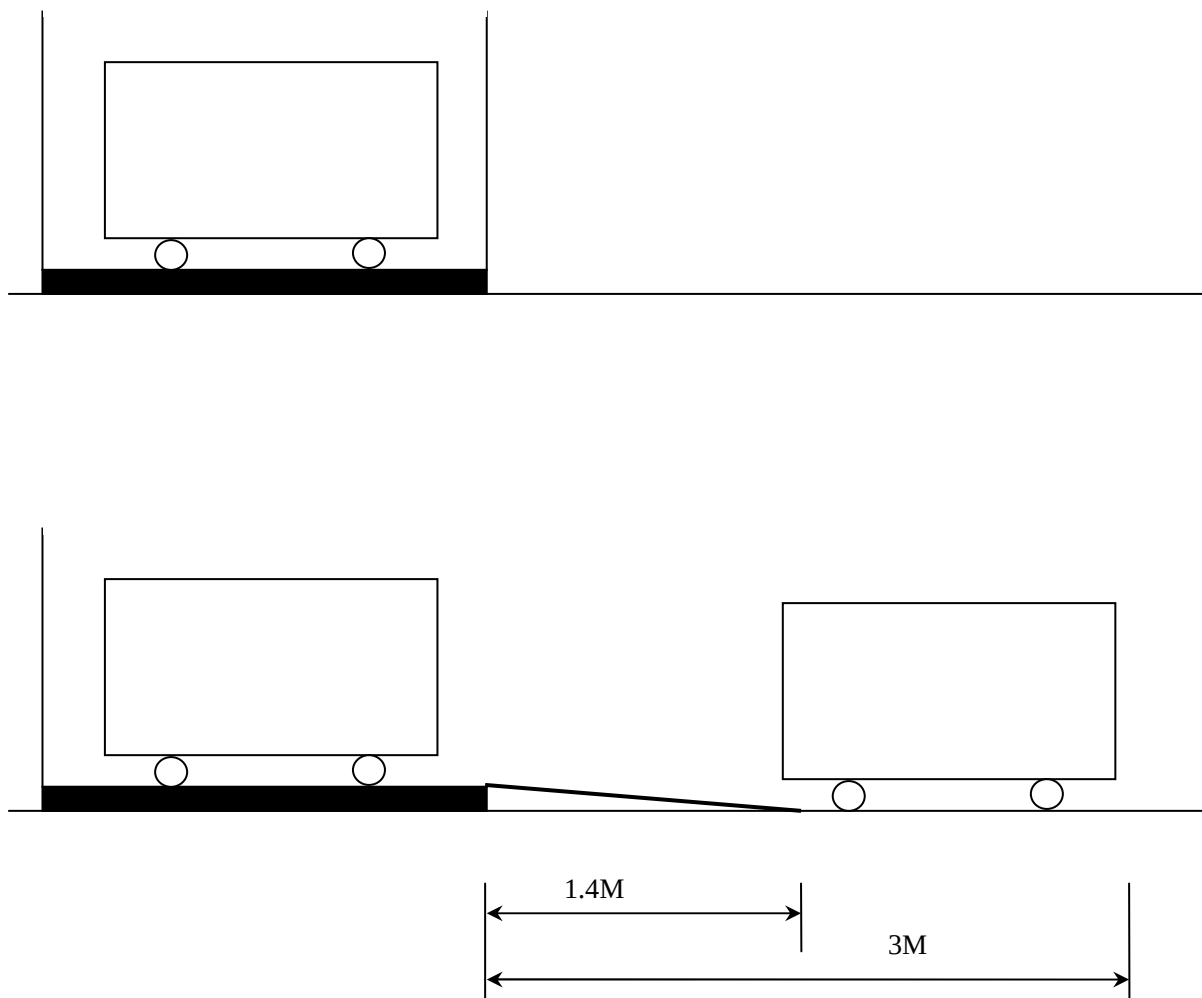
1. Before unpacking, check the packing list and check if the following items are included: chemistry analyzer, accessory kit, high-concentration waste tank, concentrated wash solution, ISE module accessory kit(optional), and water supply module.

2. Unload the chemistry analyzer from the truck. A fork lifter is recommended. Check if the Tilt Watch is red or not. Red means the instrument experienced tilting during transportation, which may cause damage or affect the performance.



3. Loosen the retaining screws on the cover of the package, remove all covers and keep 3m space in front of the slope cover (1.4m for the slope cover and 1.6m for the instrument).





- 4.** Rotate the four anchors with big wrench. Make them down to the platform to support the machine.



- 5.** Removing the 4 mounting plates by socket spanner and extender.



- 6.** There is a piece of slope plate fixed on the side board. Loosen 2 fixed screws (see left picture) and take it off from the side plate (see right picture).



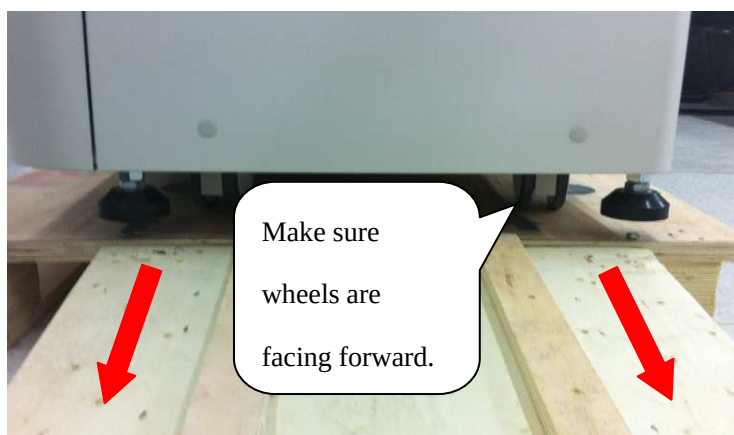
- 7.** Connect the slope plate to the baseboard (see left picture) and then fix them by two long screws to prevent sliding when moving the instrument to the floor (see right picture).



- 8.** Rotate the four anchors up so the machine is on its wheels and ready to move.



- 9.** Push chemistry analyzer via slop cover to the desired place.

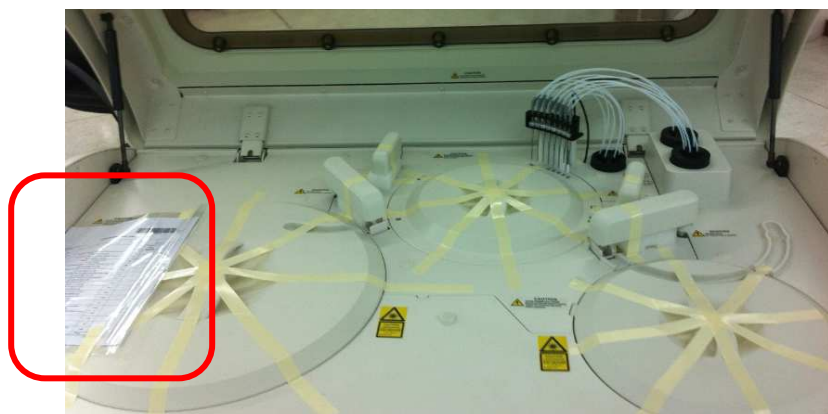


- 10.** Keep all crate covers and screws for future packing use.

3.3 Installation

3.3.1 Keep the Parameter List Paper

Open the machine cover, the parameter list paper is attached on the top panel. Ask the doctor to keep it in a safe place or just keep it with you. In addition, you are highly recommended to make a soft copy of the list, for example, take a photo or make a scanning.



3.3.2 Install Probes

Notice: Please keep all the adhesive tape except the tape of three probes. It can prevent screws dropping into machine during the probes installation.

1. Wash hand to eliminate the static electric. Wear anti-static gloves is highly recommended.
2. Take out the probes from the accessory kit. Acupuncture needles are attached outside the box (see left picture). Probe washer is inside the bag with the probe (see right picture).



Acupuncture needles



Probe washer

- 3.** Remove the probe cover then take out the retaining screw with spring for probe.



- 4.** Wet the washer and put it into the probe connector. Wet washer should stick to the probe. Put some paper cover the wash well to prevent the washer from dropping into the wash well.



- 5.** Insert the probe downwards into the big hole on the probe arm



- 6.** Connect the tube connector to the probe and then tighten it.



- 7.** Fix the grounding wire of the probe (Only reagent probe has grounding wire)



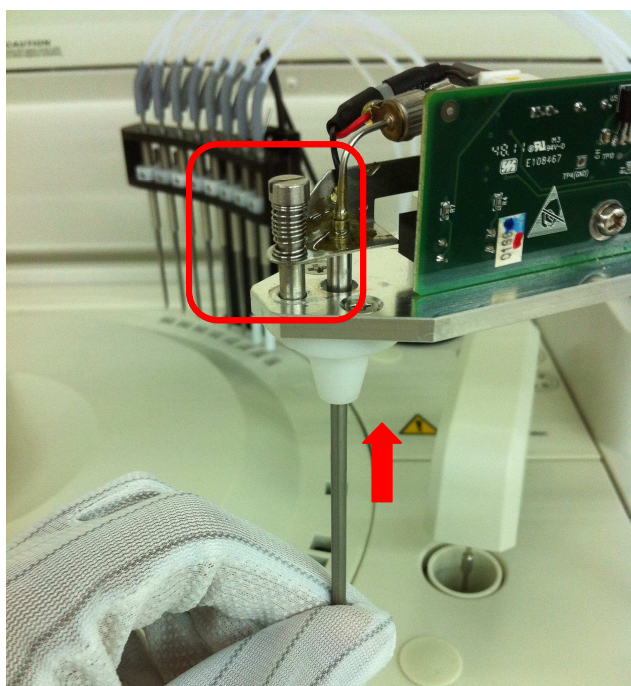
- 8.** Connect the probe cable connector to the liquid level detection board.



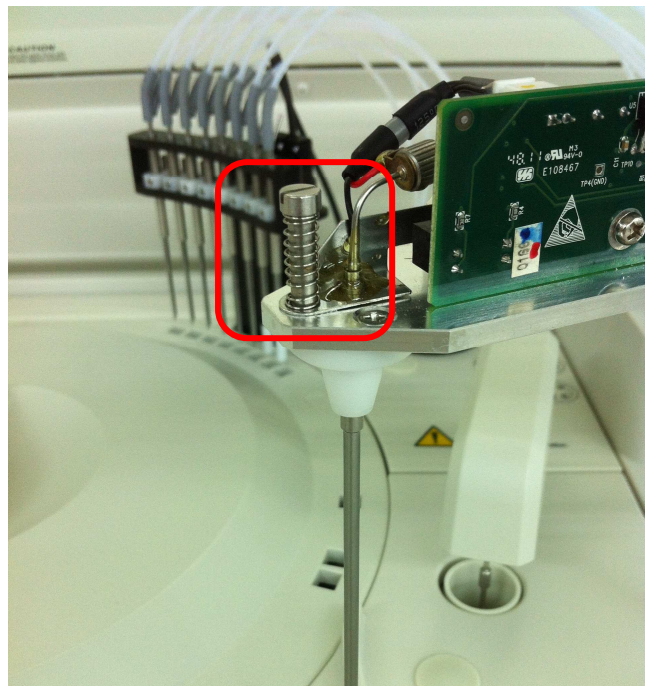
- 9.** Tighten the retaining screw with spring (see left picture). Pay attention that the spring direction and make the thread opening face downwards (see right picture).



- 10.** Pinch the probe by the part near the probe arm. Push the probe upwards (see left picture) and then release it (see right picture) to check if the probe moves downward automatically. If it does, proceed to the next step. If not, check if the spring is clamped or fixed too tightly.



Push the probe upwards

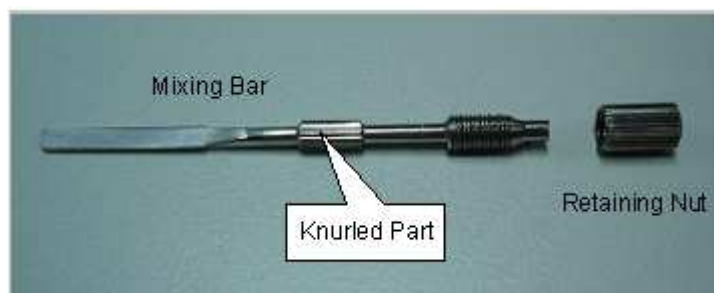


Release the probe

- 11.** DO NOT install the probe cover now. You need check for the board status and leakage later.

3.3.3 Install Mixers

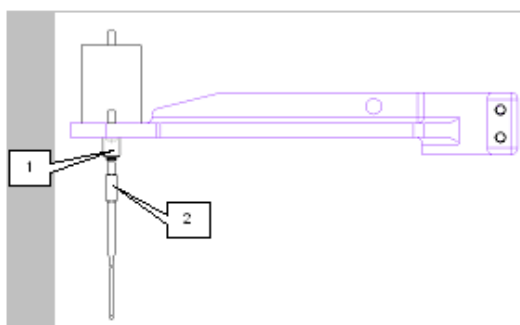
1. Rotate the bar arm to the highest position for convenient operation.
2. Take out the mixing bar and retaining nut from the box. Pinch the knurled part of the bar and take out it.



3. Pinch the new mixing bar by the knurled part and align it to the bigger hole of the retaining nut and gently screw it into the nut until the end of the bar is in line with the smaller hole end of the nut.



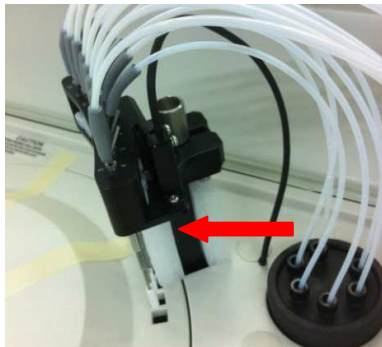
4. Pinch the mixing bar by the knurled part 2 and align the hole of the nut 1 to the axis on the bar arm, then push the bar upward in the direction of the axis until it cannot proceed.



5. Tighten the nut by screwing clockwise with the other hand. Check whether the bar is vertical to the bar arm. If not, remove the bar and re-install it.
6. Reinstall mixer cover.

3.3.4 Remove all Adhesive Tape and Foam

After probe and mixer installation, now it is safe to remove all other adhesive tape and protection foam. **DO NOT** forget the foam under the wash station.

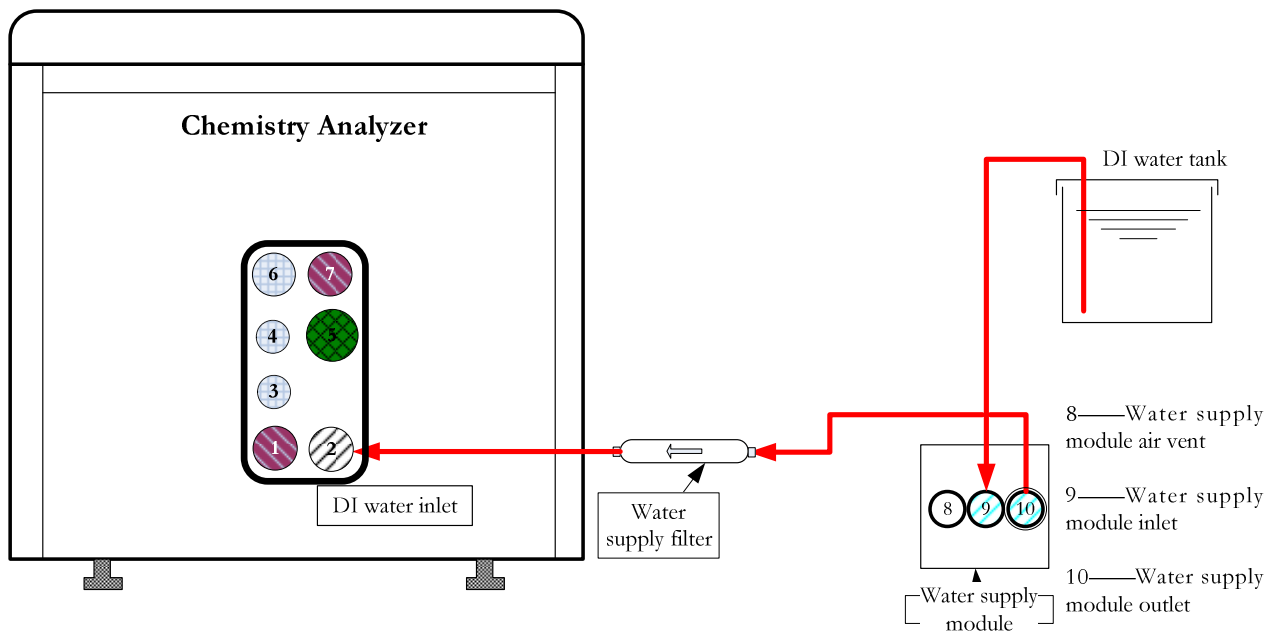


3.4 Fluidic Connection

The picture below shows the correspondence relation of the liquid connection:



3.4.1 Connect BS-480 with Water Supply Module



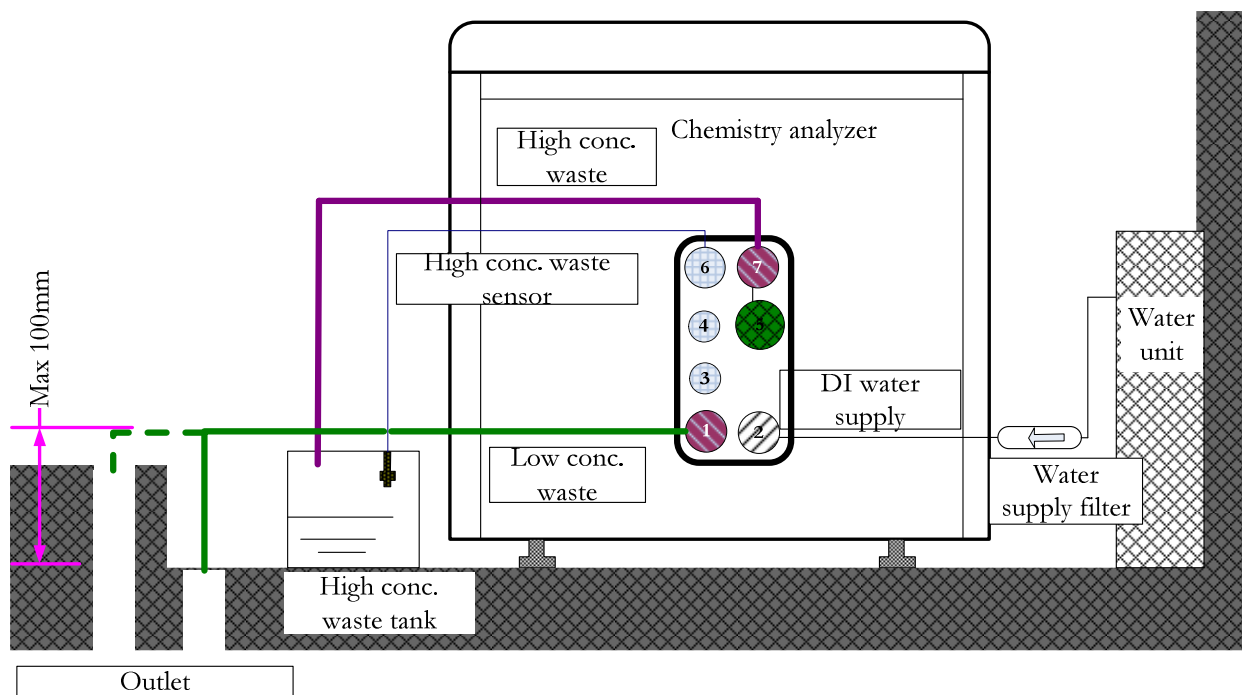
1. Use the PU tube in the accessory kit to connect the analyzer with the water supply module. Install the water supply module between the deionized water tank and the analyzer inlet. Install the water inlet filter between the water supply module outlet and the analyzer inlet, and make sure that the filter will not be influenced by external force and is convenient for replacement.
2. Make sure that the arrow on the water inlet filter shell that indicates the water flow points at the analyzer inlet.



3. Ensure that the water inlet tube between the deionized water tank and the analyzer inlet is less than 10 meters long. When installing the tube, prevent it from being bent, twisted, or pressed, and keep it away from any sharp-edged objects or others that may scratch the tube.
4. Connect the air vent of the water supply module to the sewer and keep the tube less than 10m long.

5. If the tube is not installed correctly, remove it, disconnect the part that has contacted the connectors, and then reinstall the tube properly.

3.4.2 Connect BS-480 with Waste Drainage Facilities



1. Connect the high-concentration waste tube and sensor (high-concentration waste tank cap assembly) to the high-concentration waste tank.



- 2.** Connect high-concentration waste tube and sensor to the machine.

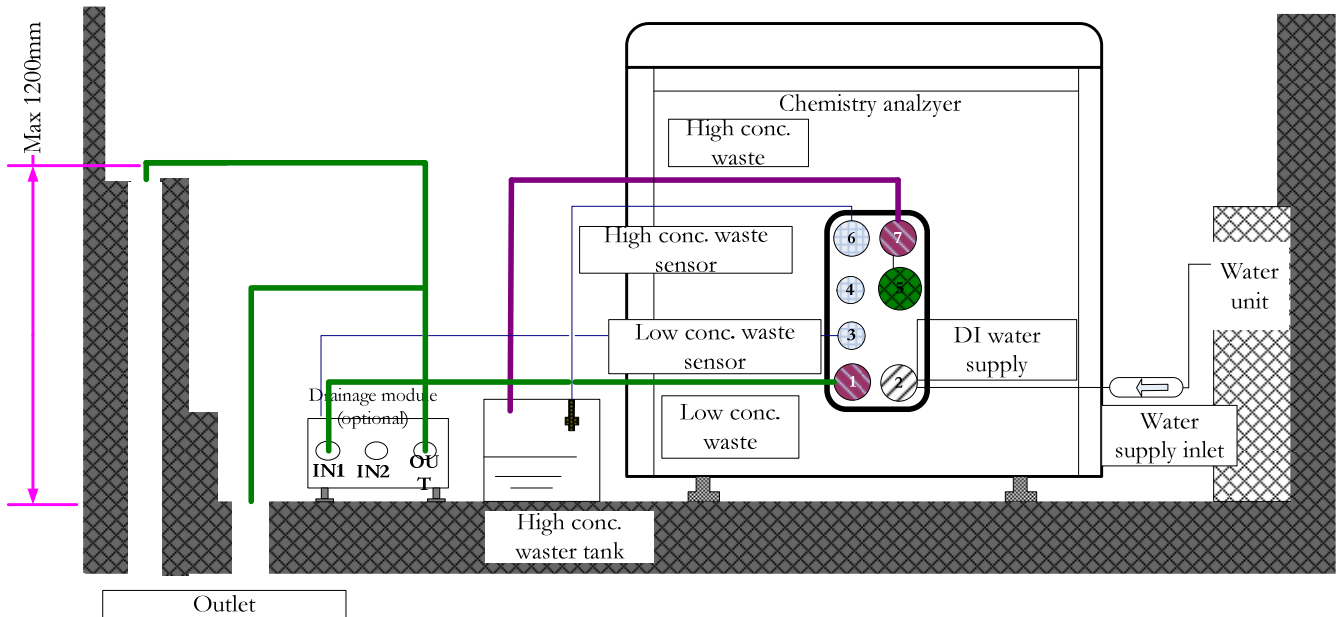


- 3.** Insert the low-concentration waste tube into the sewer, and make sure that the tube is less than 5m long. To prevent leaking, fix low concentration tube properly with the tube clamps in the accessory kit.



- 4.** Make sure that the low-concentration waste outlet is no less than 50mm wide and less than 100mm high.
- 5.** When installing the tubes, prevent them from being bent, twisted, or pressed, and keep them away from any sharp-edged objects or others that may scratch the tubes.
- 6.** Ensure that all electric connectors are higher than and away from fluidic parts and prevent fluidic connectors from facing electric components.

3.4.3 Connect BS-480 with Drainage Module (Optional)



In case the sewer outlet is higher than 100mm, it is necessary to install a drainage module to ensure a smooth waste drain. However, even with the drainage module, the sewer outlet should NOT higher than 1,200mm.

- 1.** Connect the high-/low-concentration waste tubes separately.
- 2.** Connect the high-concentration waste tube and sensor (high-concentration waste tank cap assembly) to the high-concentration waste tank.
- 3.** Connect the low-concentration waste outlet to the inlet of the drainage module with a tube no more than 5m long; and then connect the low-concentration waste sensor to the drainage module.
- 4.** Connect the drainage module outlet to the low-concentration waste outlet with a tube no more than 10m long.
- 5.** Make sure that the low-concentration waste outlet is no less than 50mm wide and less than 1.2m high.

To prevent leaks, fix all tubes and connectors properly with the tube clamps in the accessory kit.

3.5 Install the Concentration Detergent (CD80)

1. Open the door and remove the protective foam. Then you will find the detergent cap assembly is located behind the foam.



2. Insert the cap assembly into the detergent pack and make the cap tight. The other side of the tube and sensor cable has already been connected in factory.



3.6 Install the ISE Reagent Package and Electrode (Optional)

1. Take off the adhesive tape for ISE reagent package.



2. Take out the protective foam slowly and carefully to prevent damaging the ISE wand.



- 3.** Connect the wand with ISE reagent package then put the package in.



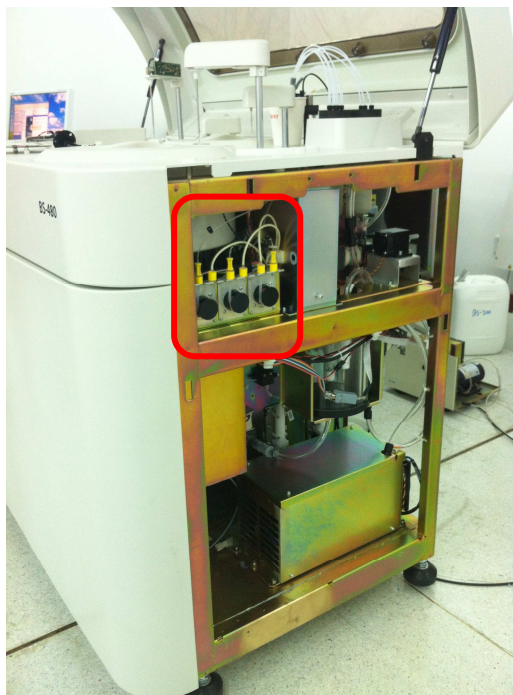
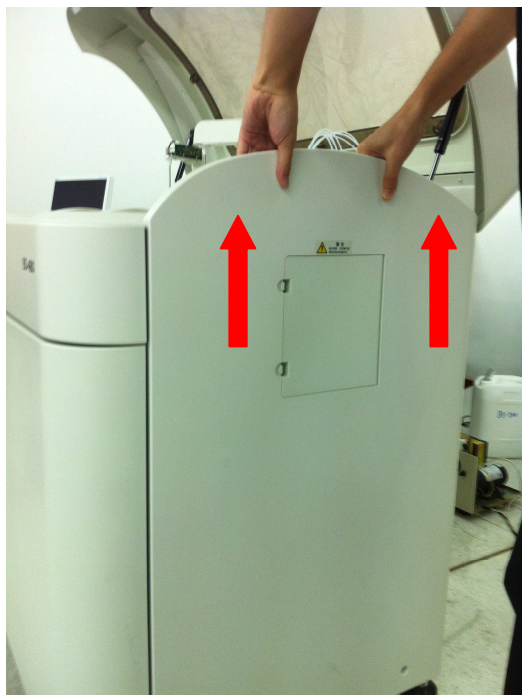
- 4.** Remove the ISE rubber cap on the top panel.



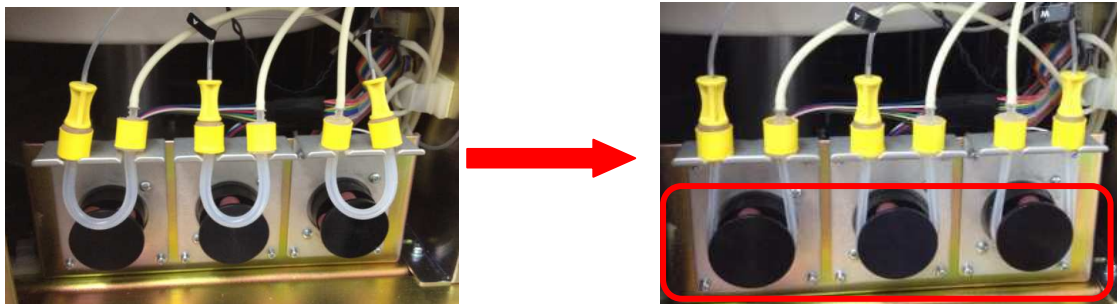
- 5.** Remove the two rubber caps at the bottom of the right panel. Then remove the two screws inside.



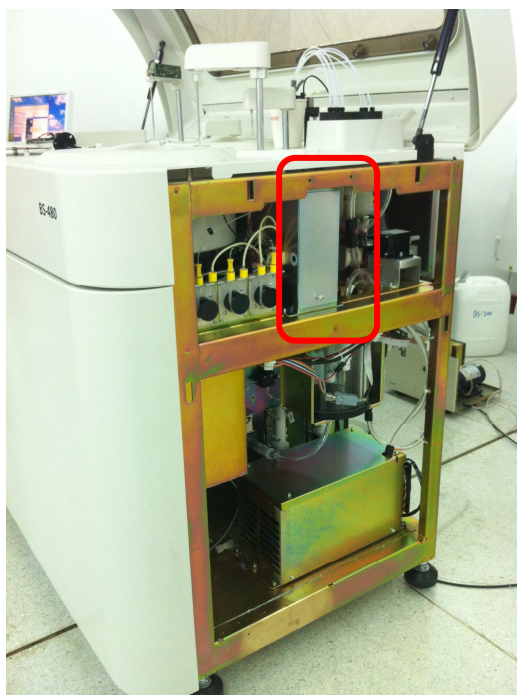
- 6.** Lift the panel to remove (see left picture). Then you can find ISE pumps inside (see right picture).



- 7.** Drag and put the peristaltic tube on the pump wheel according to the pictures below. DO NOT reverse the tubing directions.



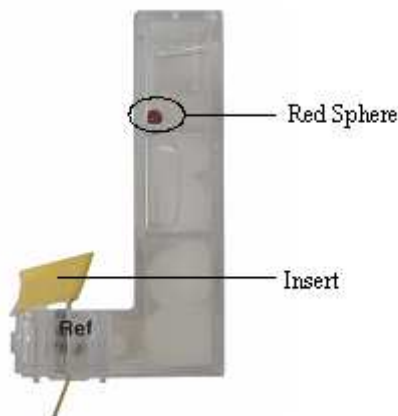
- 8.** Find the ISE module on the right side of the pumps.



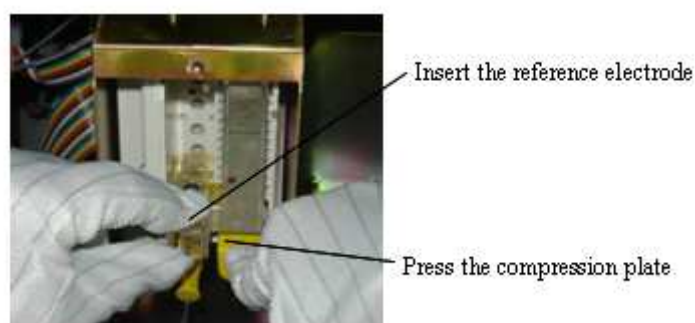
- 9.** Unscrew the captive screw of the shielding box (see left picture) to show the electrodes housing (see right picture).



- 10.** The reference electrode is the first to be installed. Open the electrode from its protective packaging and remove the insert from the lumen of the reference electrode. Make sure that the red sphere in reference electrode floats on the top of the internal solution in the reservoir.

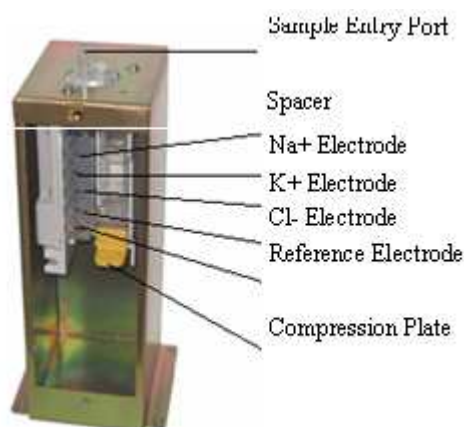


- 11.** Place the reference electrode inside the housing by pressing down the compression plate and push it straight against the back of the housing. Release the compression plate and ensure the electrode cannot be easily moved.



- 12.** Remove the Chloride electrode from its protective packaging and place it in the ISE module housing in the same way as the reference electrode.
- 13.** Repeat the process for the Potassium electrode.
- 14.** Repeat the process for the Sodium electrode.
- 15.** Repeat the process for Spacer.

- 16.** Push all the electrodes simultaneously to ensure they are in correct alignment. All the electrodes for spacer, Na, K and Cl are of the same size and shape. Connection pins at the rear of each electrode are different and ensure that the electrodes are inserted in the correct order. So if one of the electrodes cannot be easily pushed into the housing, check the electrode first then repeat the installation process.



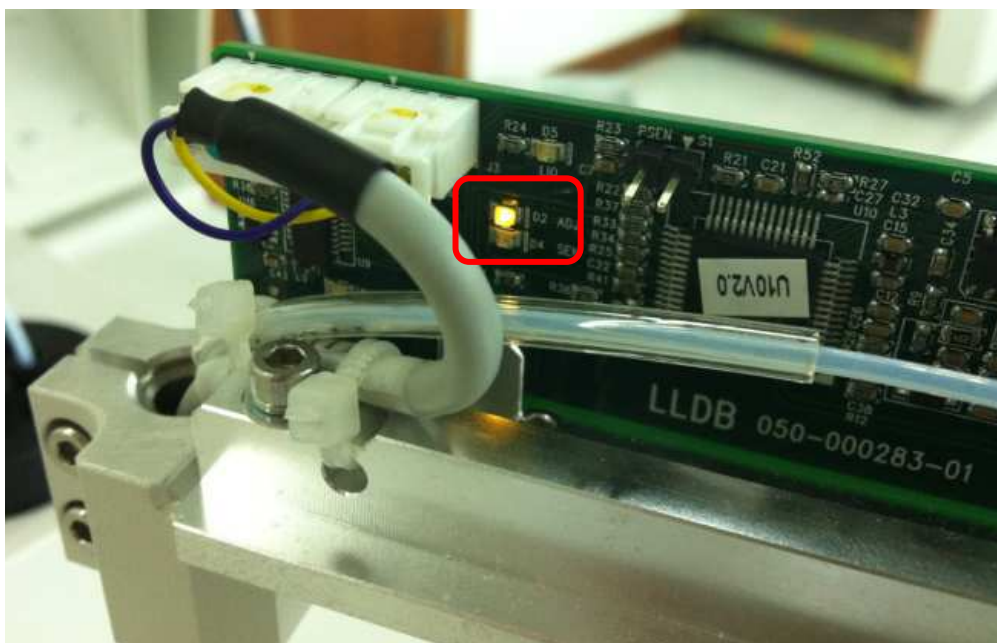
- 17.** Install the cover back to the shielding box by tightening the screws.
- 18.** Reinstall the right panel of the machine.

3.7 Cable Connection

- 1.** Install the UPS according to its official installation guide. The UPS should be properly grounded.
- 2.** Connect UPS to the analyzer, computer, printer, monitor, water unit (and water supply module). Do not switch on the power before finishing all connections.
- 3.** Connect the computer to the analyzer with the serial port cable

3.8 Powering On

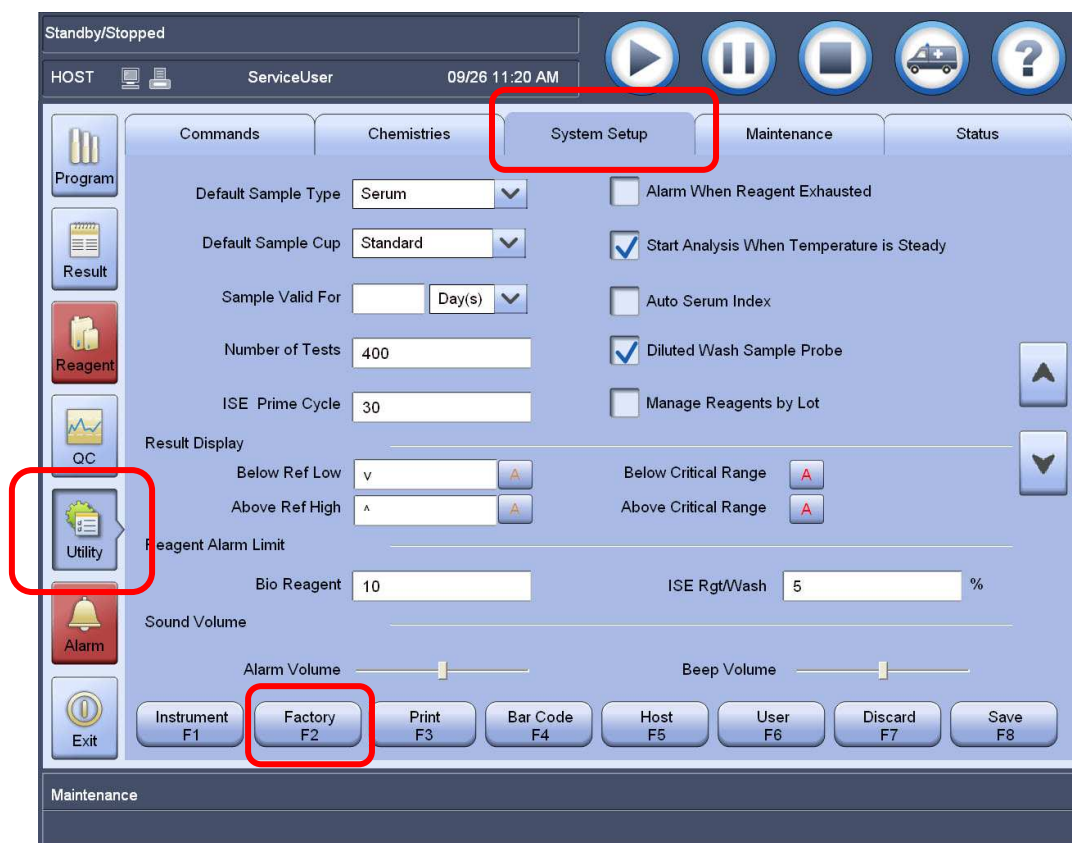
1. Turn on the power switch of the water supply module and the water drainage module (optional).
2. Turn on the main switch and analyzing unit switch of BS-480. The water inlet valve is turned on automatically. Check if there is water coming into the water tank. Normally the internal water tank can be filled full in 20-30 minutes. You can proceed with next steps when water refilling is in process.
3. Check the liquid level detection boards of sample and reagent probe. When power is on, the LED **D2** is supposed to be on. If not, press the reset button “**ADJ**” at the rear side to activate the board.



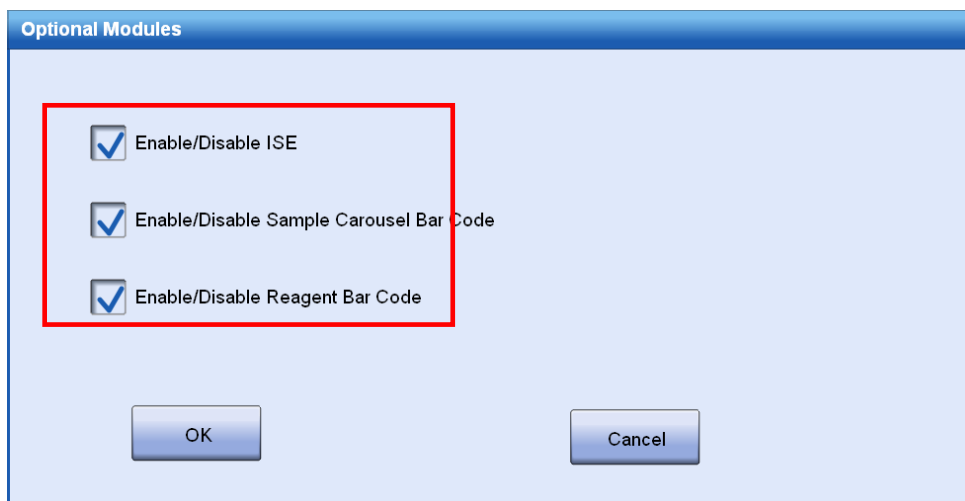
4. Turn on the computer. The BS-480 operating software is running automatically when the Windows is started.
5. For the first time running, the operation software skips the initialization procedure because the fluidic prime is not performed. Enter the username “**ServiceUser**” and password “**#BS8A#SEU**” in the login window.



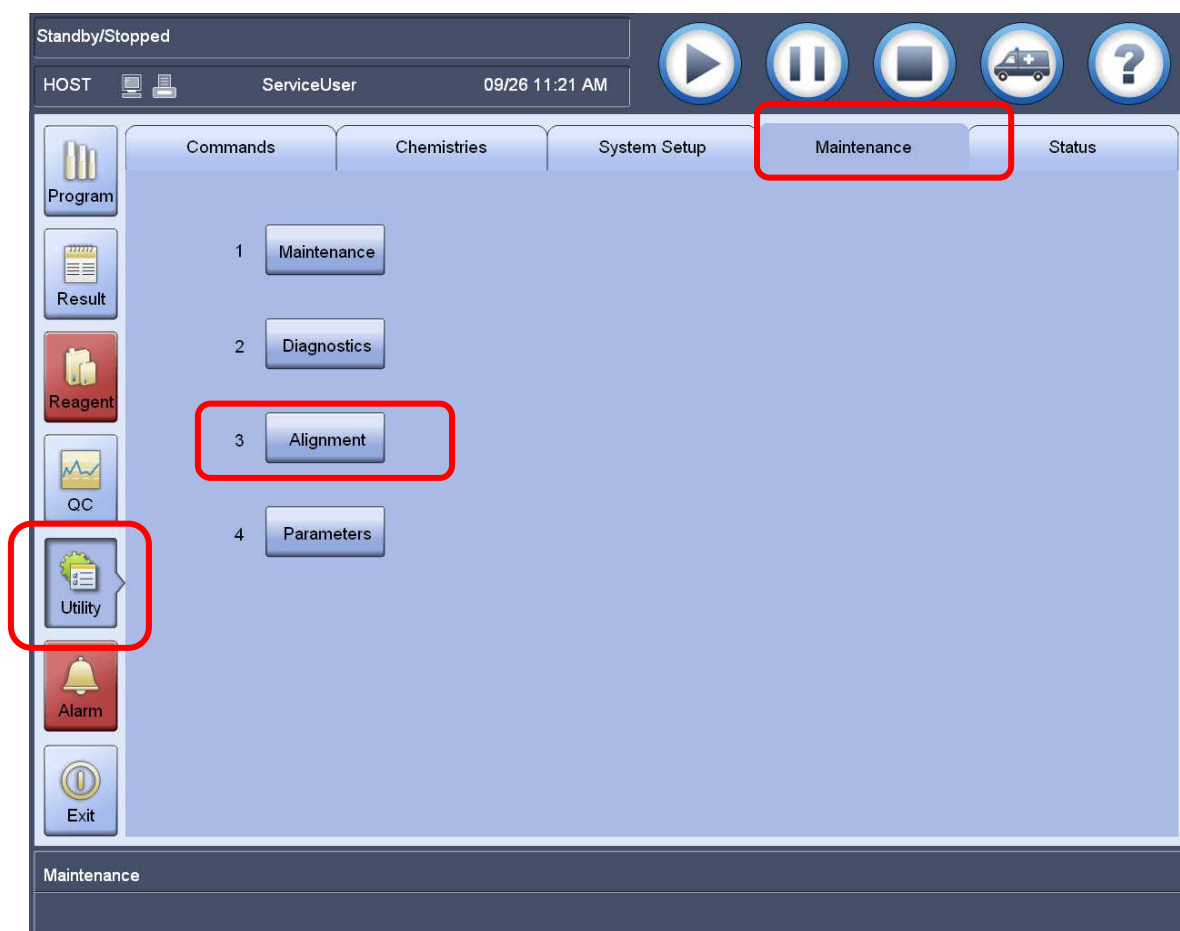
6. When the main screen is shown, select **Utility** -> **System Setup** -> **Factory F2**.



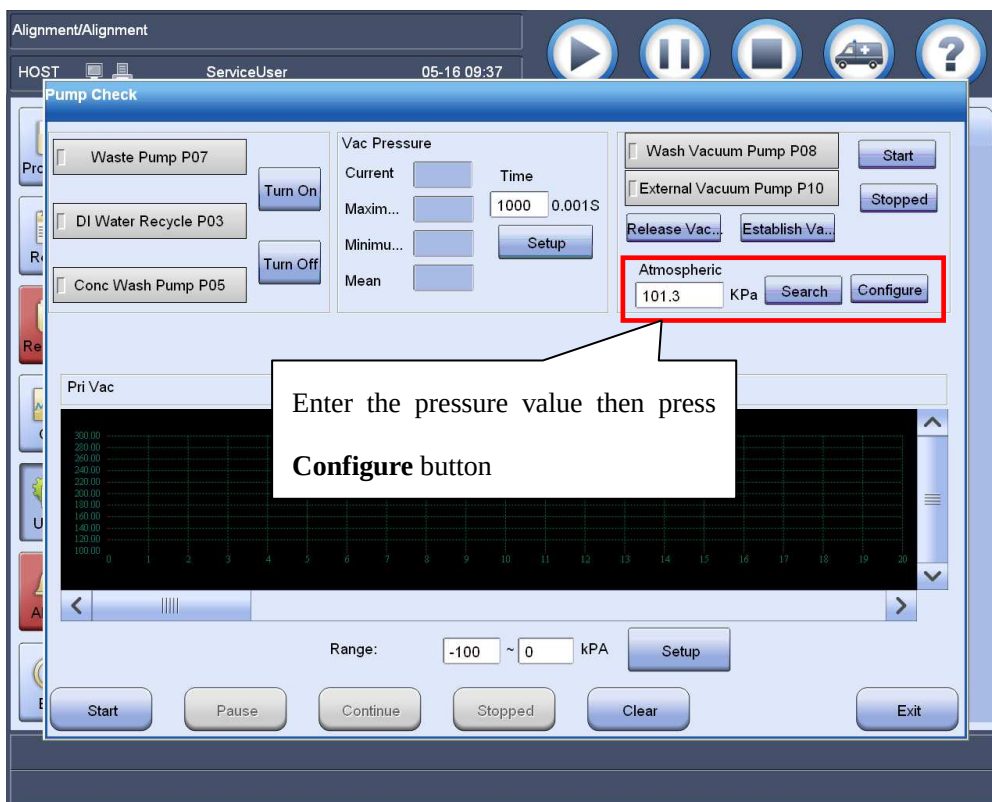
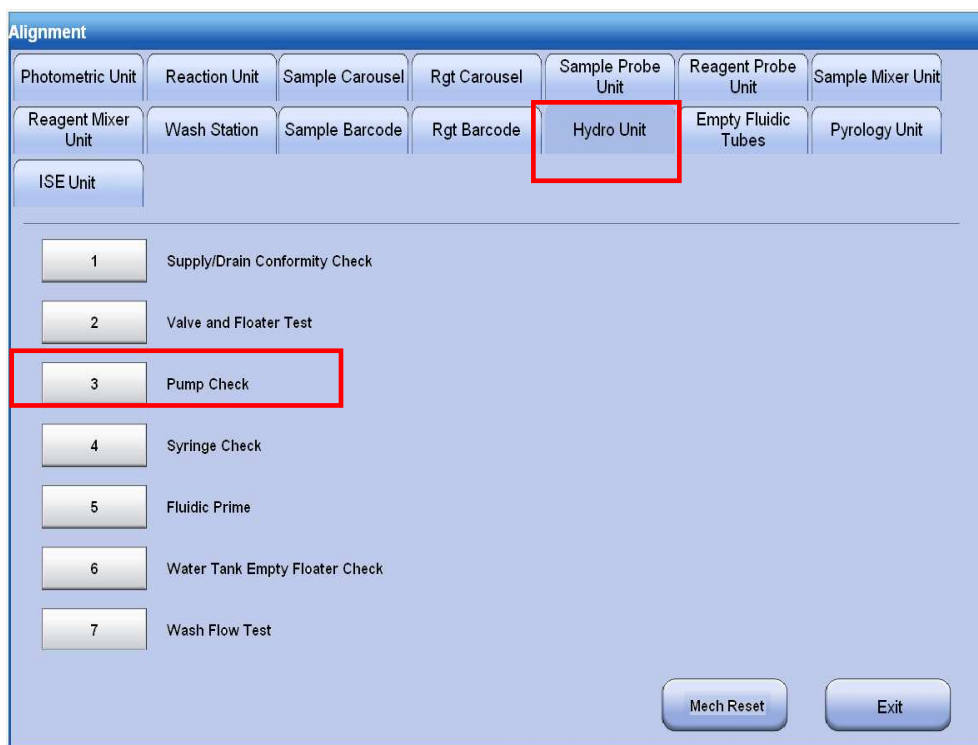
- 7.** Select **1 Optional Modules**, enable/disable these optional modules according to the machine configuration.



8. Select **Utility** -> **Maintenance** -> **Alignment**

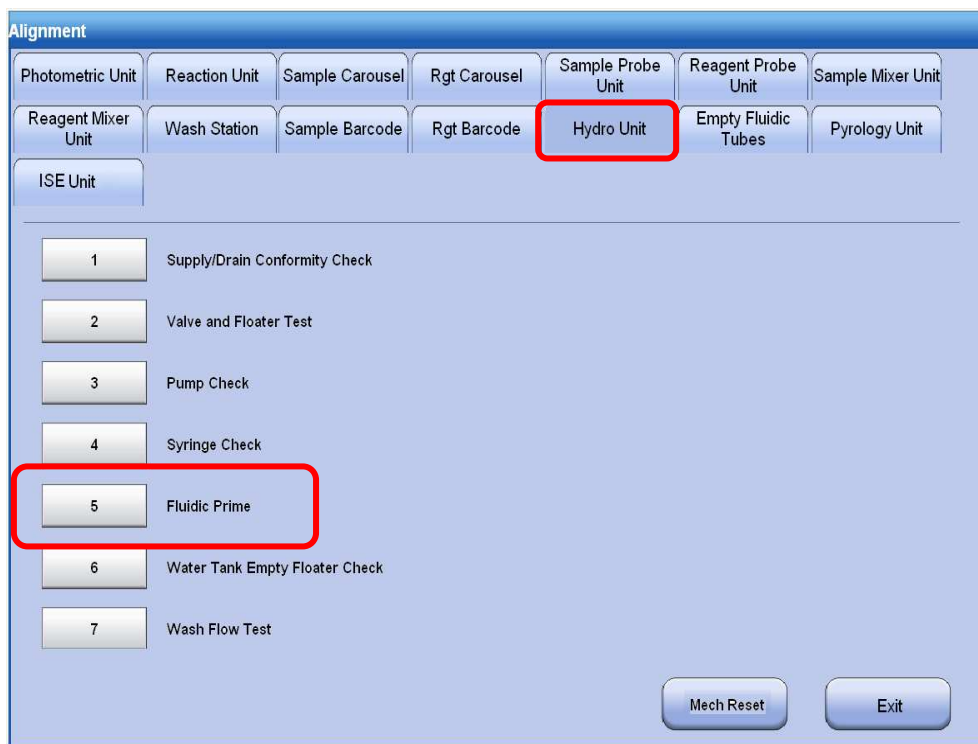


9. Select **Hydro Unit** -> **Pump Check**, enter your local atmospheric pressure and then select **Configure** to save the input. To obtain the pressure value, please first find your local altitude, then refer to Appendix on Page 73 to get corresponding atmospheric pressure value.

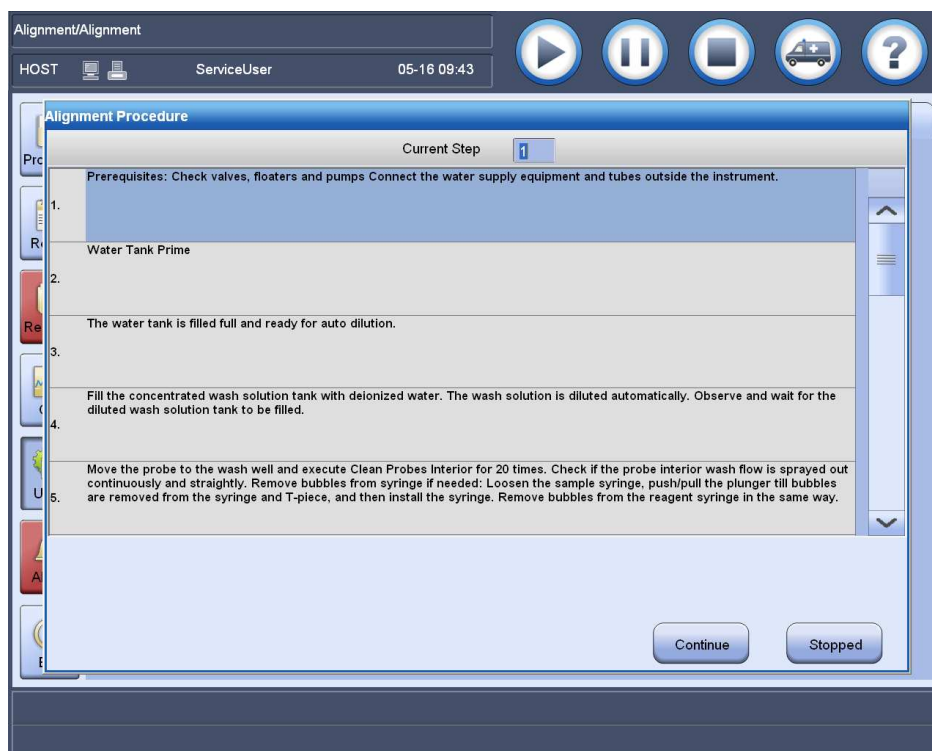


3.9 Fluidic Prime

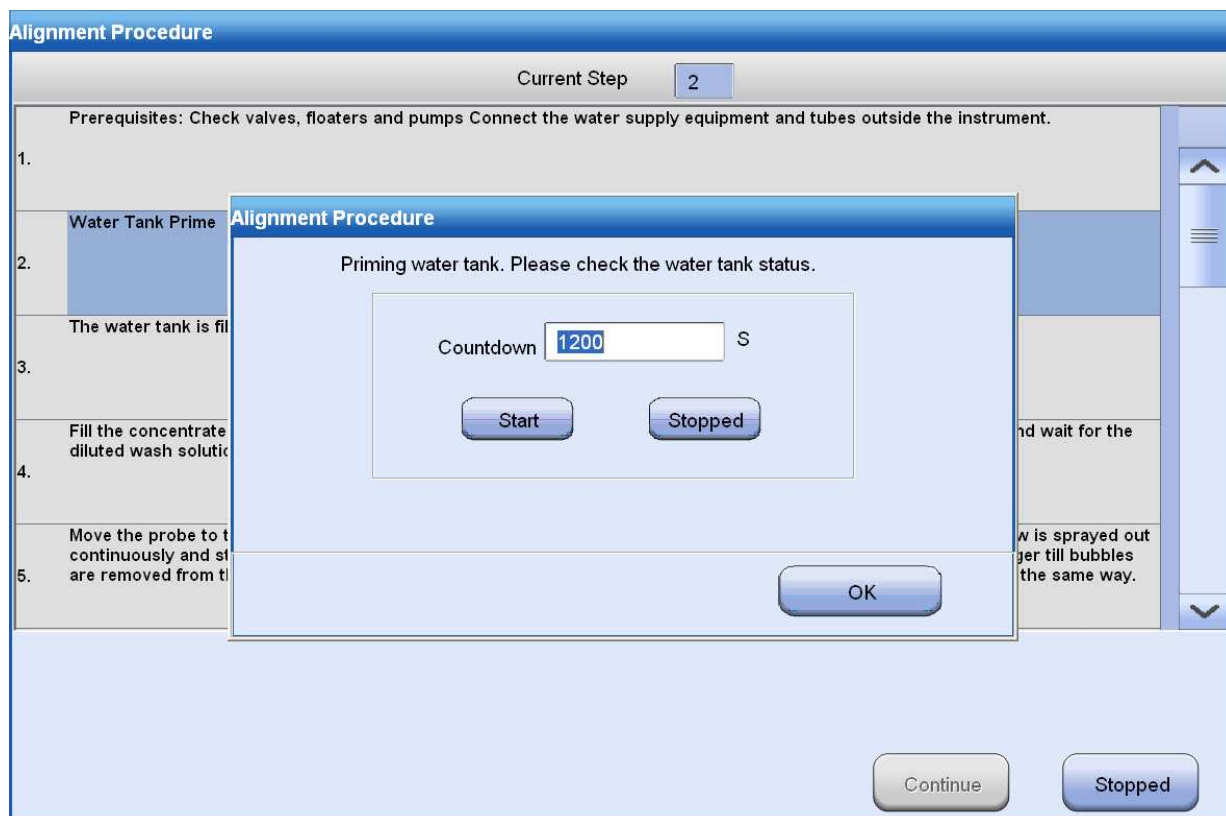
1. Select **Utility** -> **Maintenance** -> **Alignment** -> **Hydro Unit**. Select **Fluidic Prime**

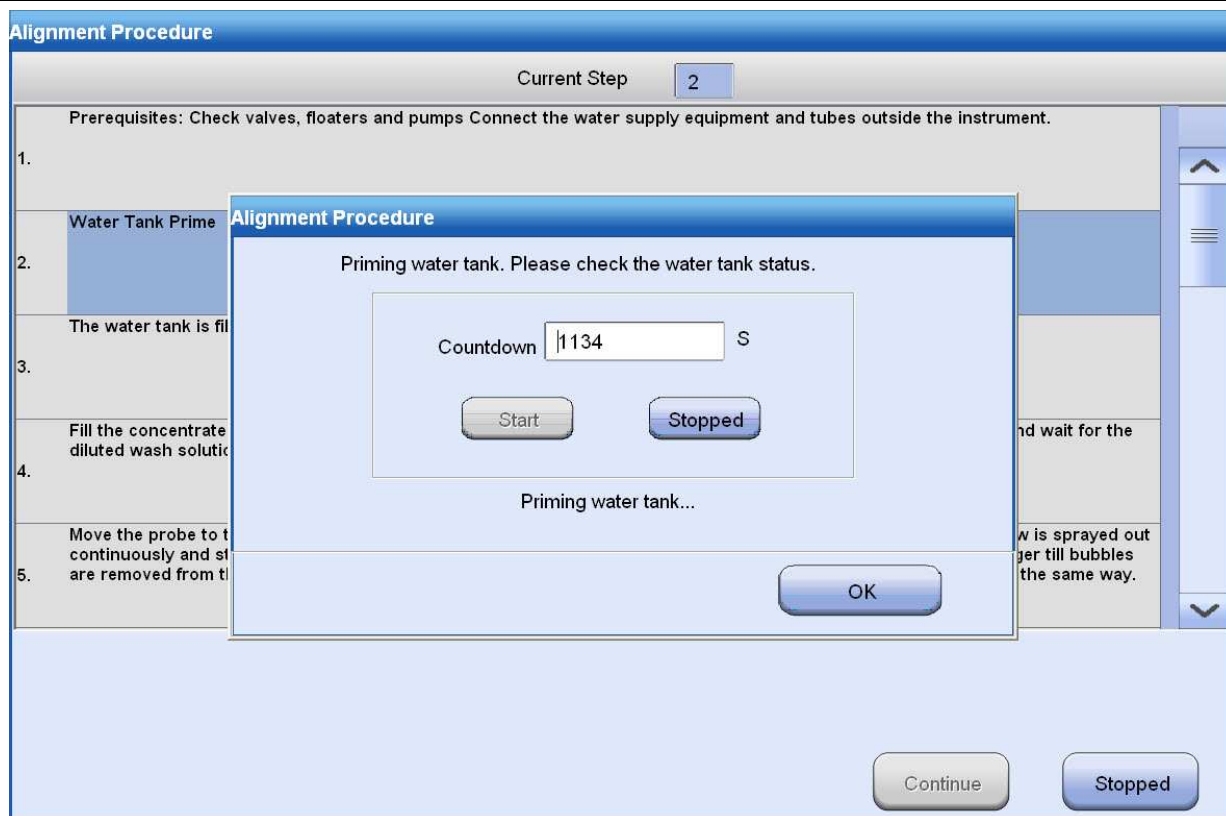


2. In step 1, please visually check any abnormal in valves/floaters/water supply module/tubings outside the machine. Select **Continue** to proceed to the next step.

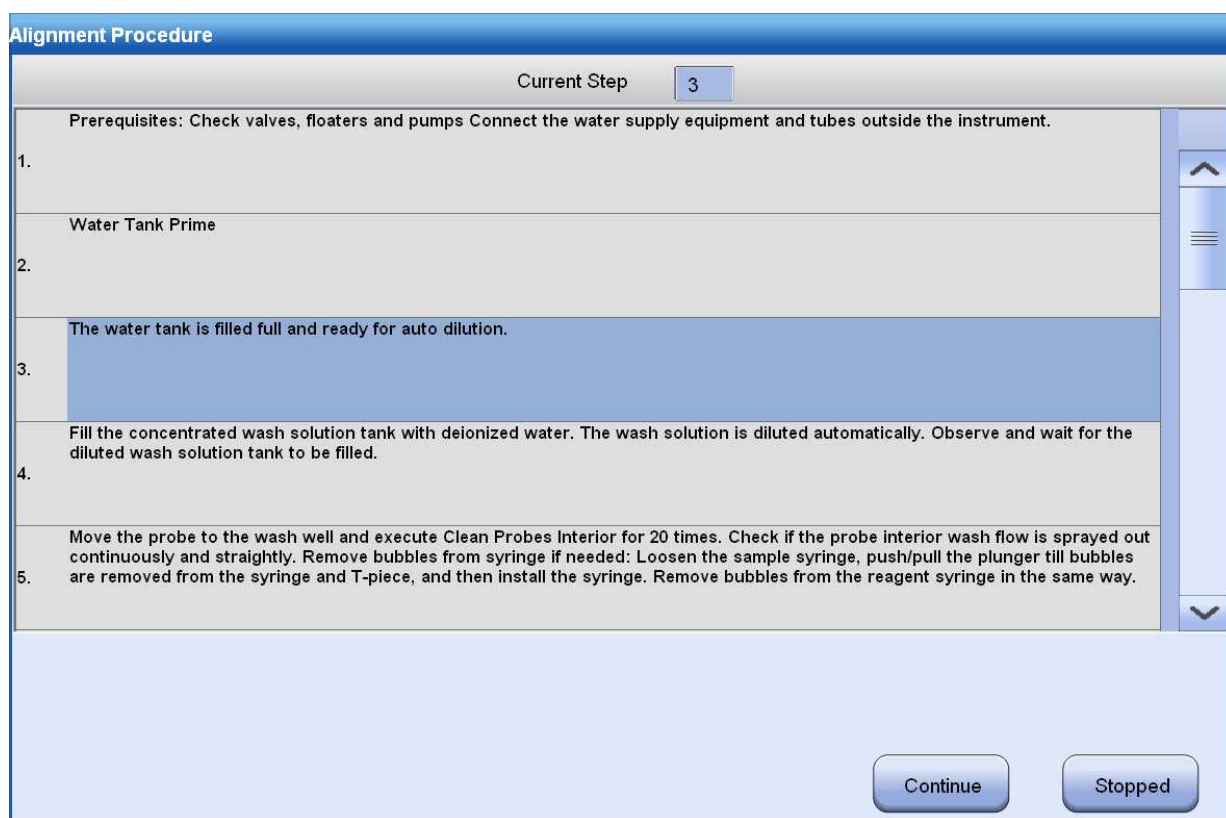


- 3.** In step 2, you can click Start then Stopped to prime the water tank. For new installation, when you turn on both switches, machine will prime the water tank automatically. You can wait in this step until water tank is full.

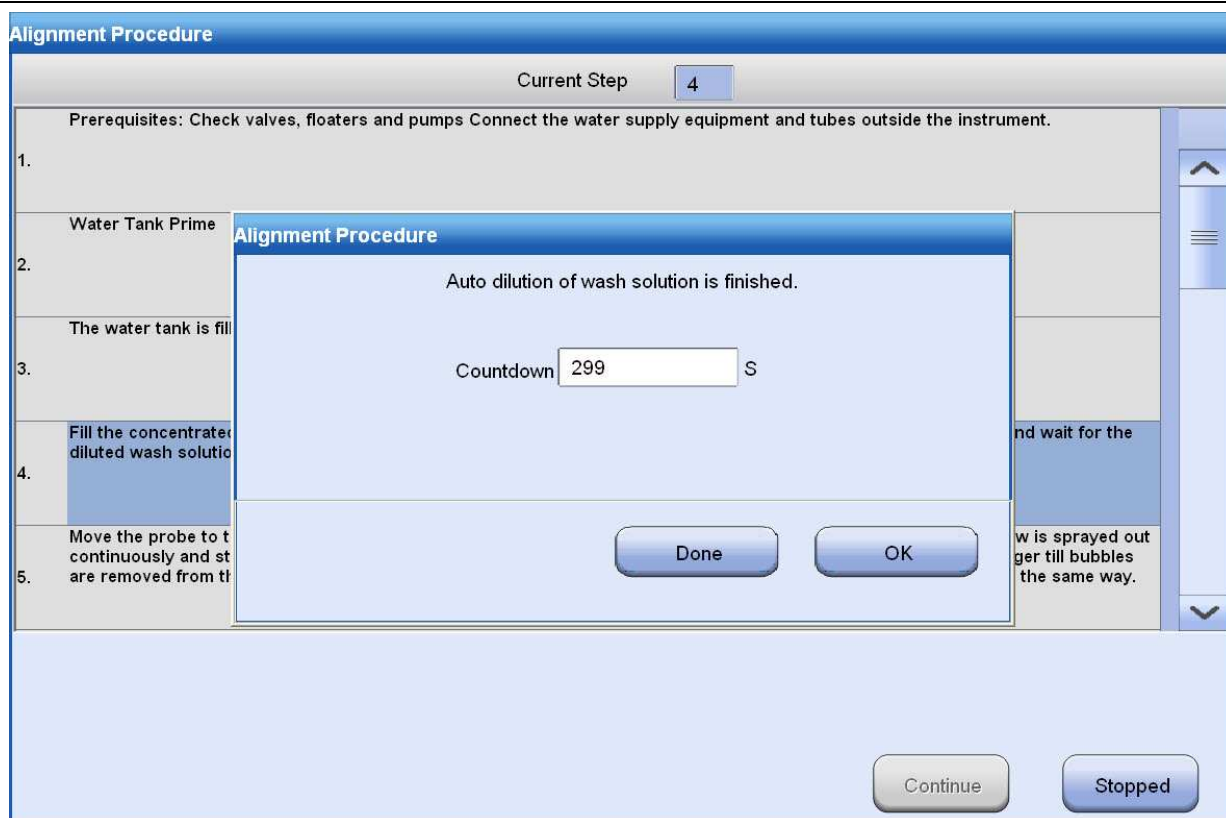




4. Step 3 indicates water tank priming is finished. Click **Continue** to proceed.



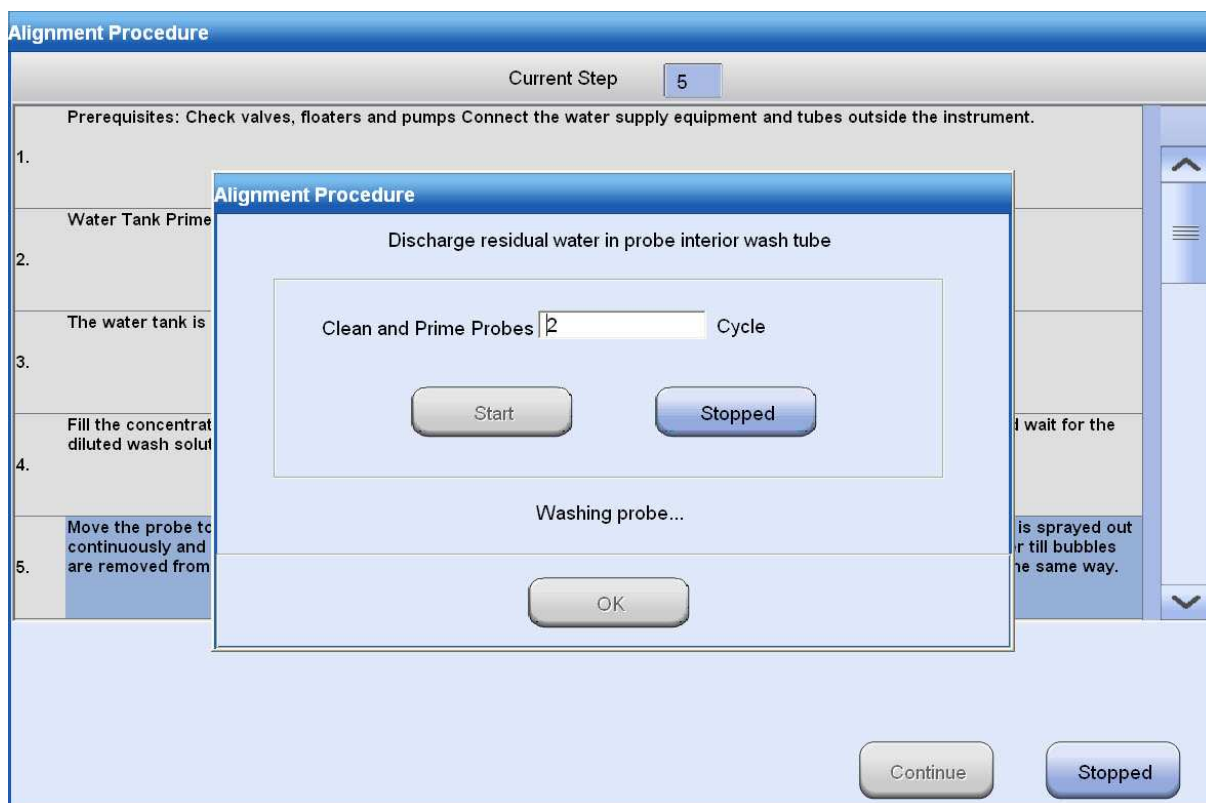
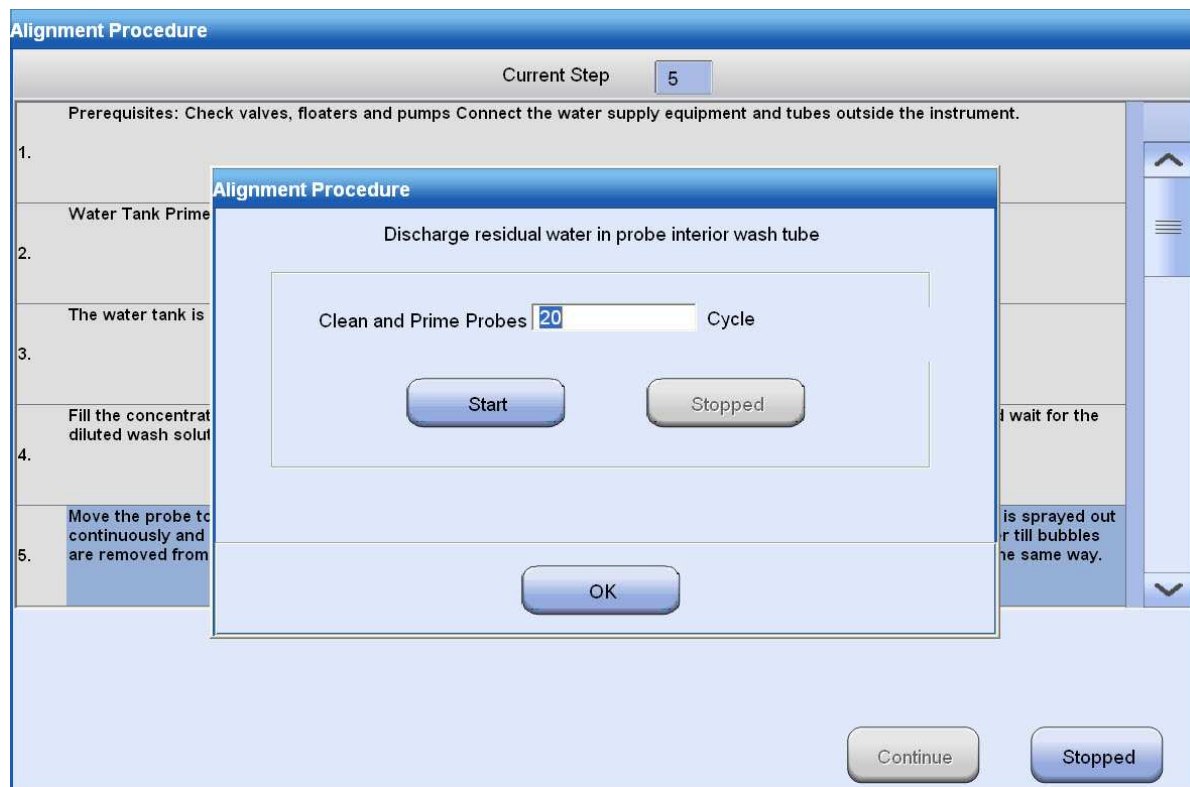
5. In step 4, select **Continue** to prime diluted wash solution tank. This step will take about 5 minutes as shown in the following figure.



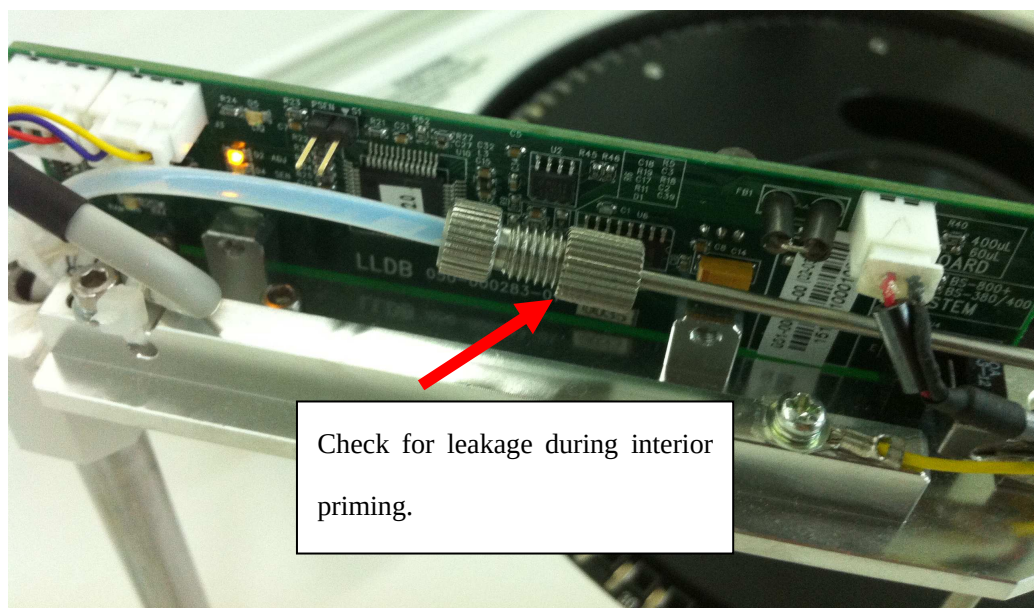
The DI water pump P03 is turned on during the process. Open the front door and observe the water pressure gauge and check if the reading is within 35-45KPa. When the auto dilution is finished (The diluted wash solution container is full or the wash solution is sufficient for the test.), select **Done** and then select **Continue** to proceed.



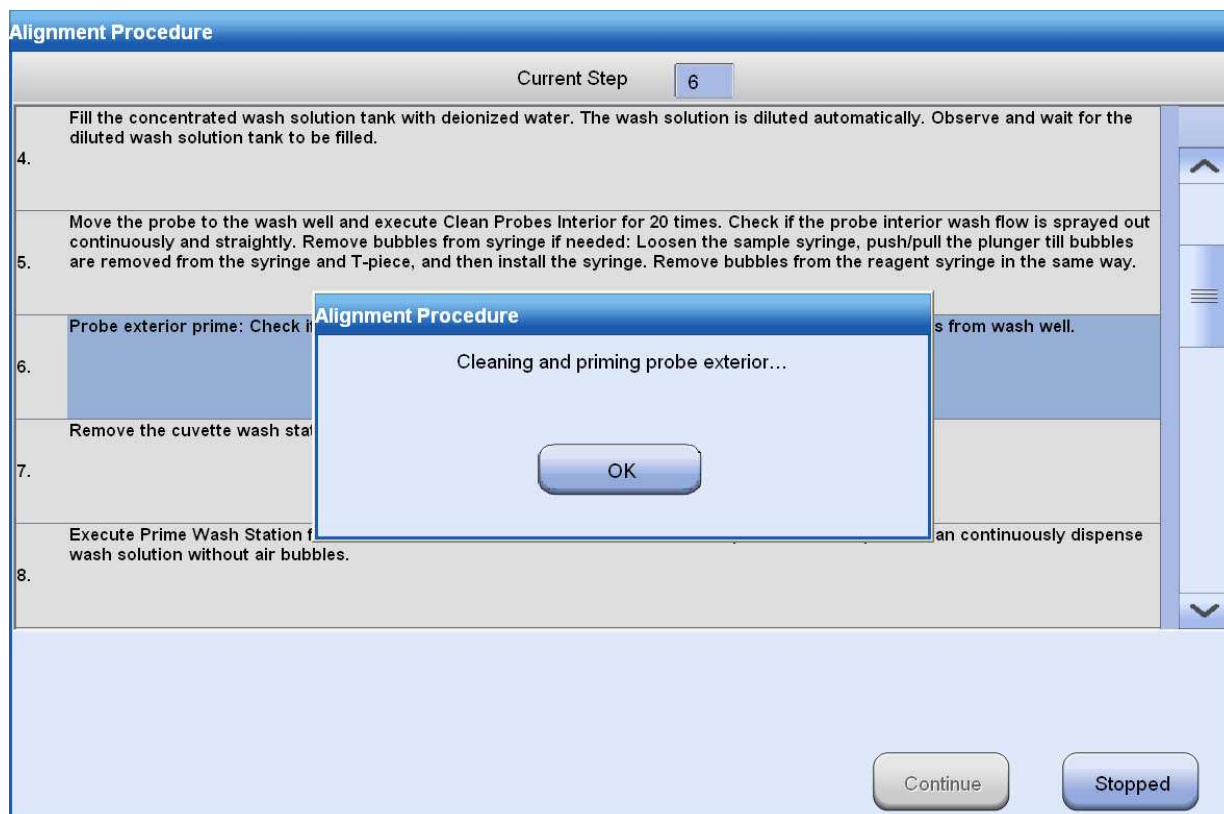
6. In step 5, select **Continue** to start prime the probes. In the popped window, keep default number “20” and click Start. Then syringes will move up and down for 20 times to prime the interior washing tubes. Please visually check if there is water spraying straight from the two probes.



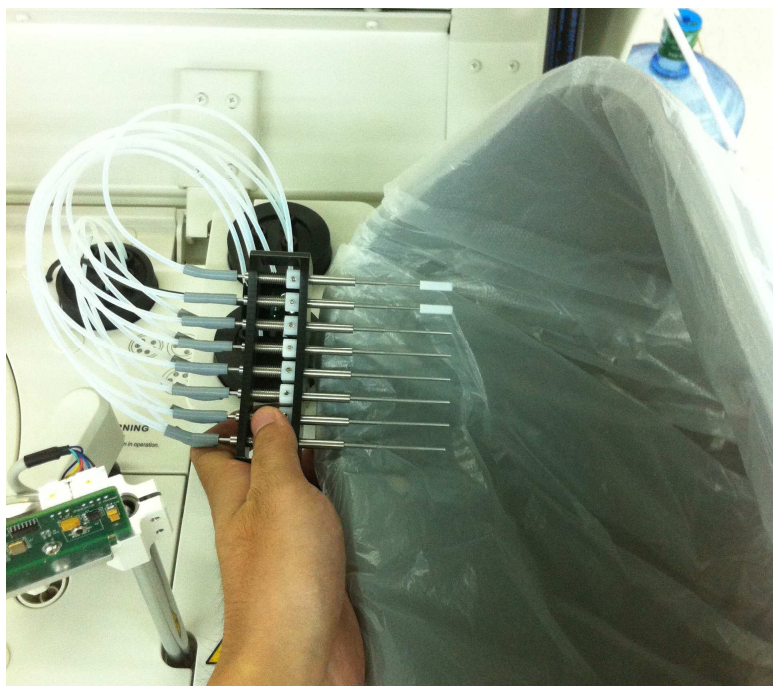
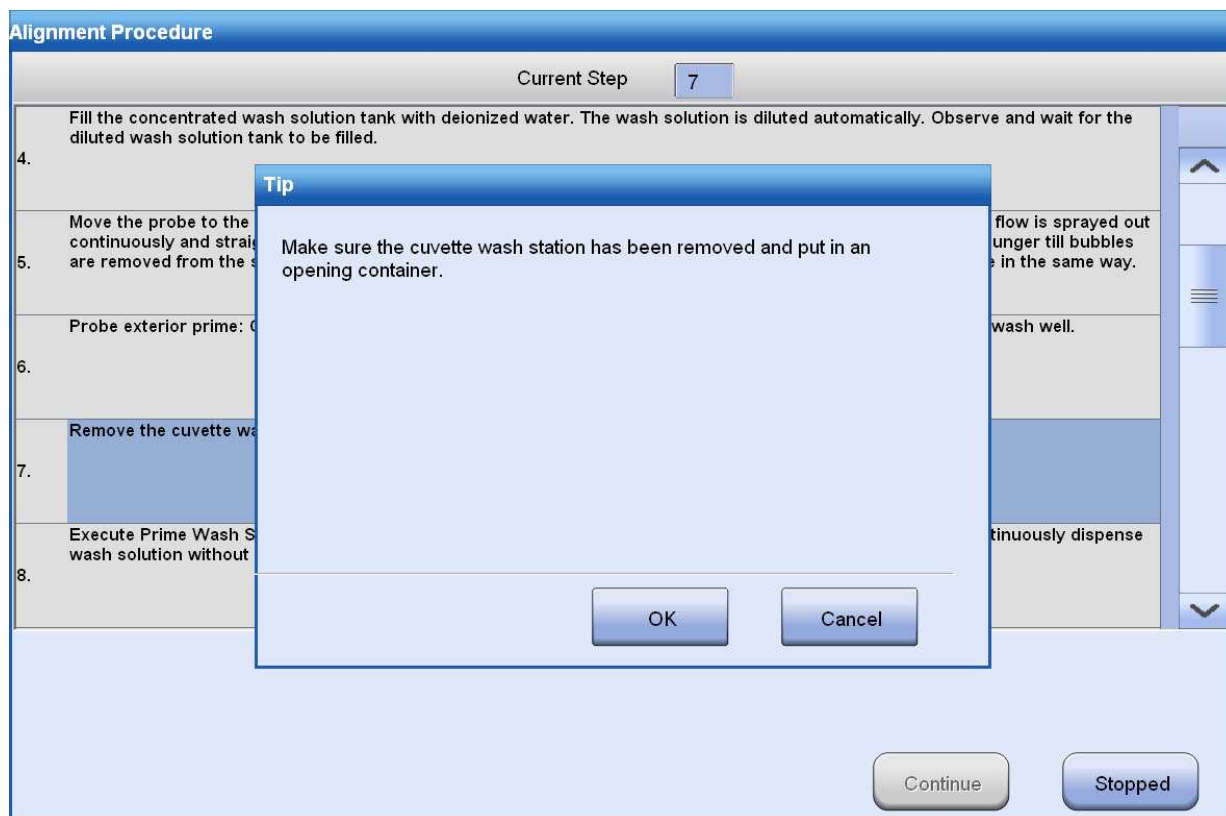
In addition, during the procedure, check if there is leakage from the probe-tube connection. If yes, please reconnect the tubings.



7. In step 6, select **Continue** to prime the exterior washing tubings. Visually check if water is spraying out continuously from all four wash wells. Select **OK** and then click **Continue** to move to the next step.



- 8.** In step 7, follow the screen instructions, remove the cuvette wash station and put it in an opening container. This step is important. If you miss this step, it will definitely have cuvette overflow. Select **OK** to proceed.



9. In step 8, select **Continue** to prime wash station phase 1 and phase 2. In popped window, keep the priming cycle as default number “100”, then click “**Start**”. Check if the two probes can spray out wash solution continuously without air bubbles, then select **Stop** and click **OK** to exit.

Alignment Procedure

Current Step: 8

7. Remove the cuvette wash station and put it in an opening container.

8. Execute Prime Wash Station for 100 times. Auto Wash Phase 1/2 Prime: Check if phase 1/2 wash probes can continuously dispense wash solution without air bubbles.

9. Auto Wash Phase 3-6 Prime: Check if phase 3-6 wash probes can continuously dispense wash solution without air bubbles.

10. Lift the 8 wash probes from the liquid level and get them ready for auto cleaning.

11. Wait for 30s until system vacuum is established, and then check if the vacuum pressure is normal.

Buttons: Continue, Stopped

Alignment Procedure

Current Step: 8

7. Remove the cuvette wash station and put it in an opening container.

8. Execute Prime Wash Station for 100 times. Auto Wash Phase 1/2 Prime: Check if phase 1/2 wash probes can continuously dispense wash solution without air bubbles.

9. Auto Wash Phase 3-6 Prime: Check if phase 3-6 wash probes can continuously dispense wash solution without air bubbles.

10. Lift the 8 wash probes from the liquid level and get them ready for auto cleaning.

11. Wait for 30s until system vacuum is established, and then check if the vacuum pressure is normal.

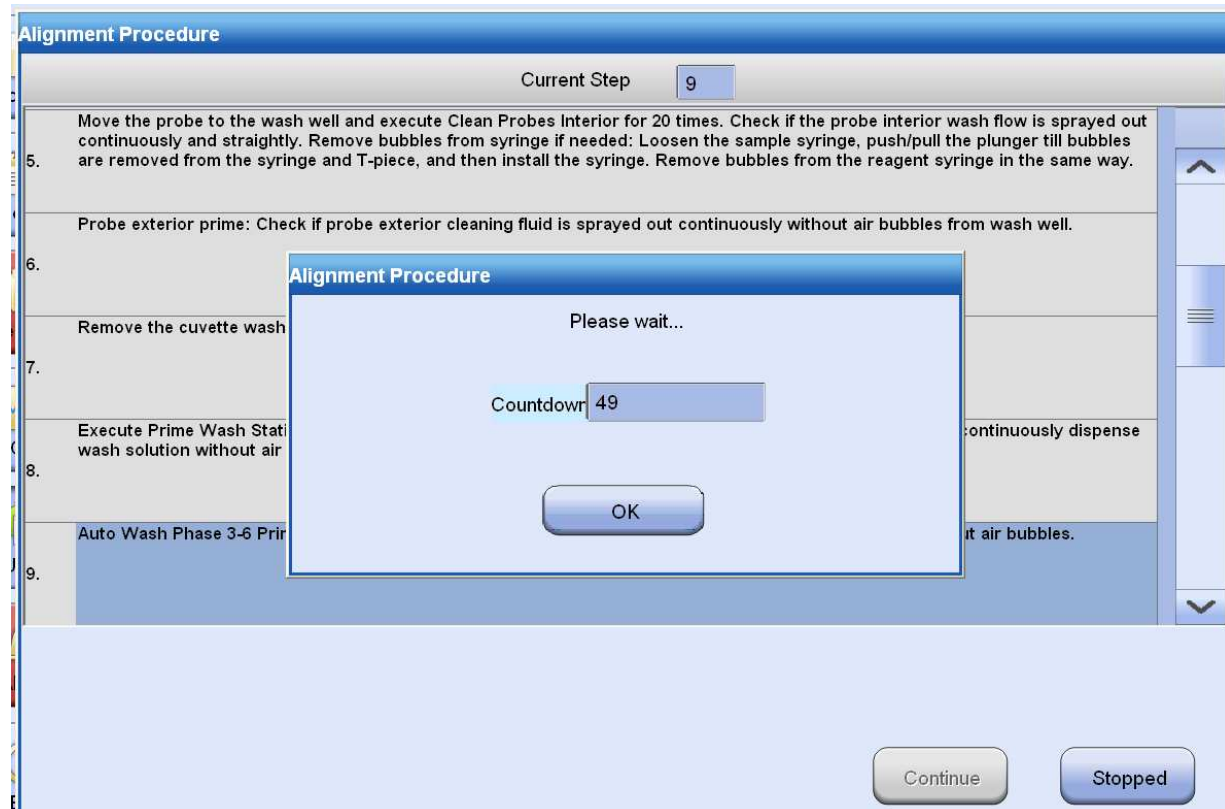
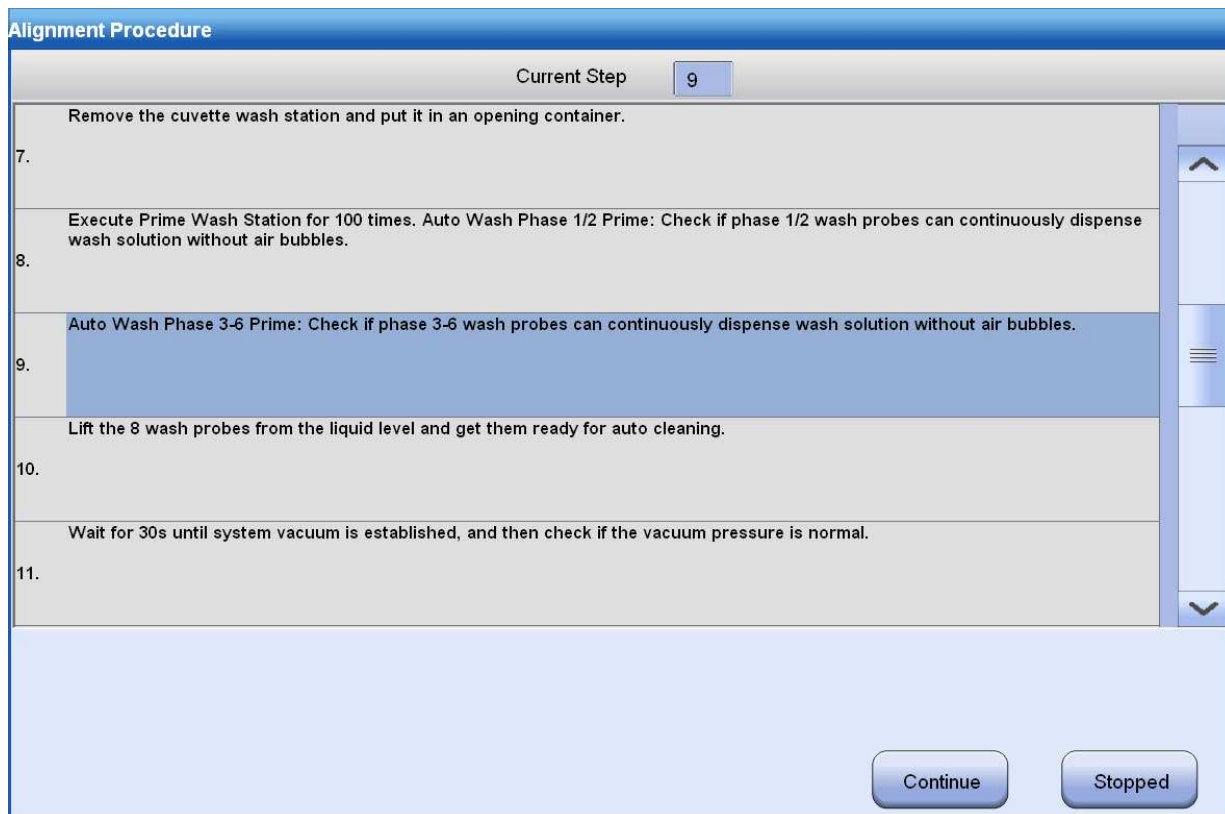
Buttons: Continue, Stopped

Alignment Procedure (Pop-up)

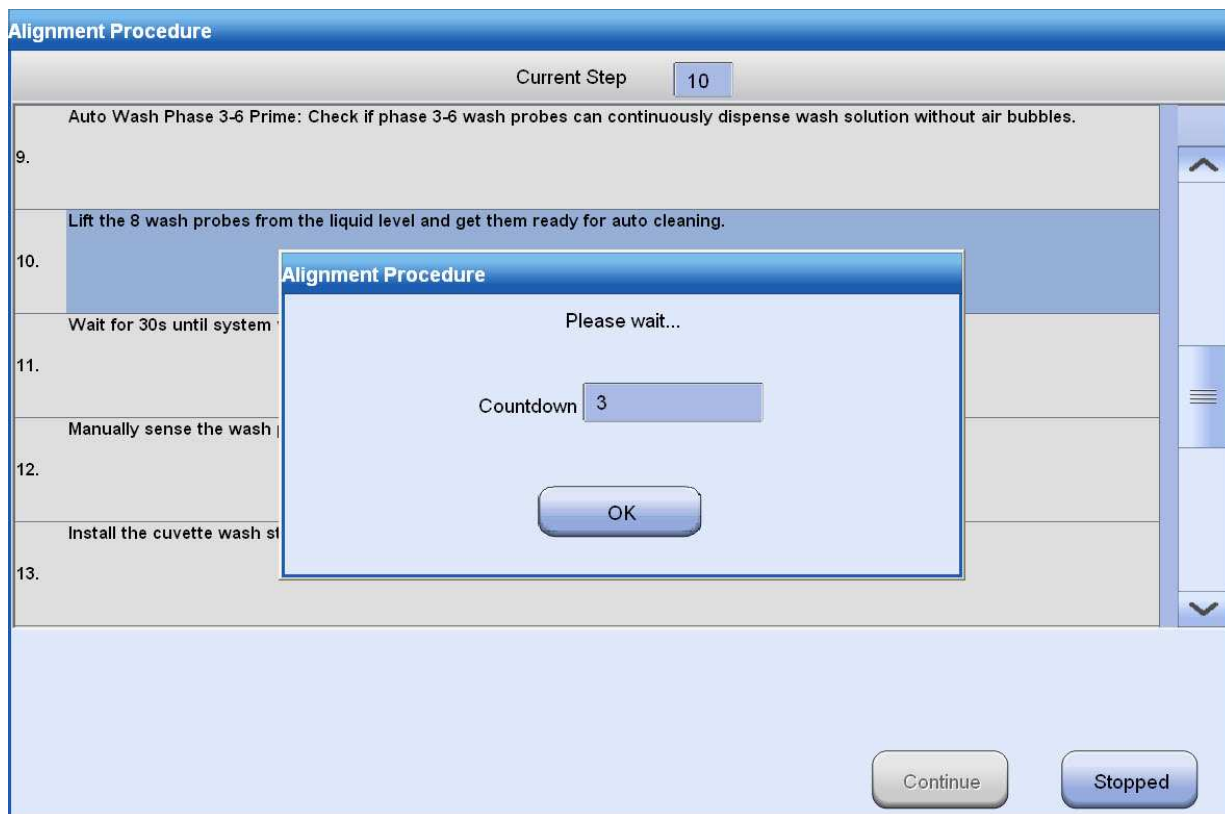
Clean and Prime Probes: 100 Cycle

Buttons: Start, Stopped, OK

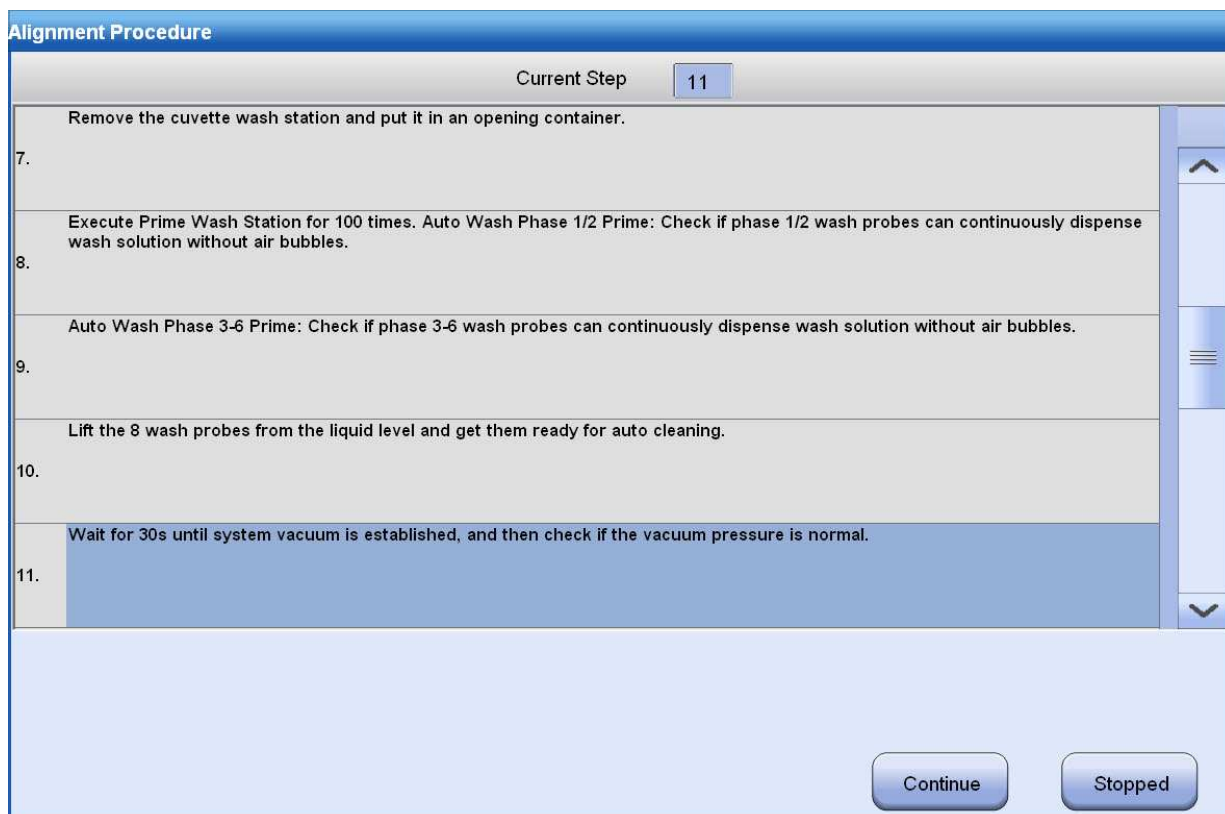
- 10.** In step 9, select **Continue** to prime wash station phase 3 to phase 6. When a countdown window is popped up, please wait and check if the four probes can spray out wash solution continuously without air bubbles, then click **OK** to exit.



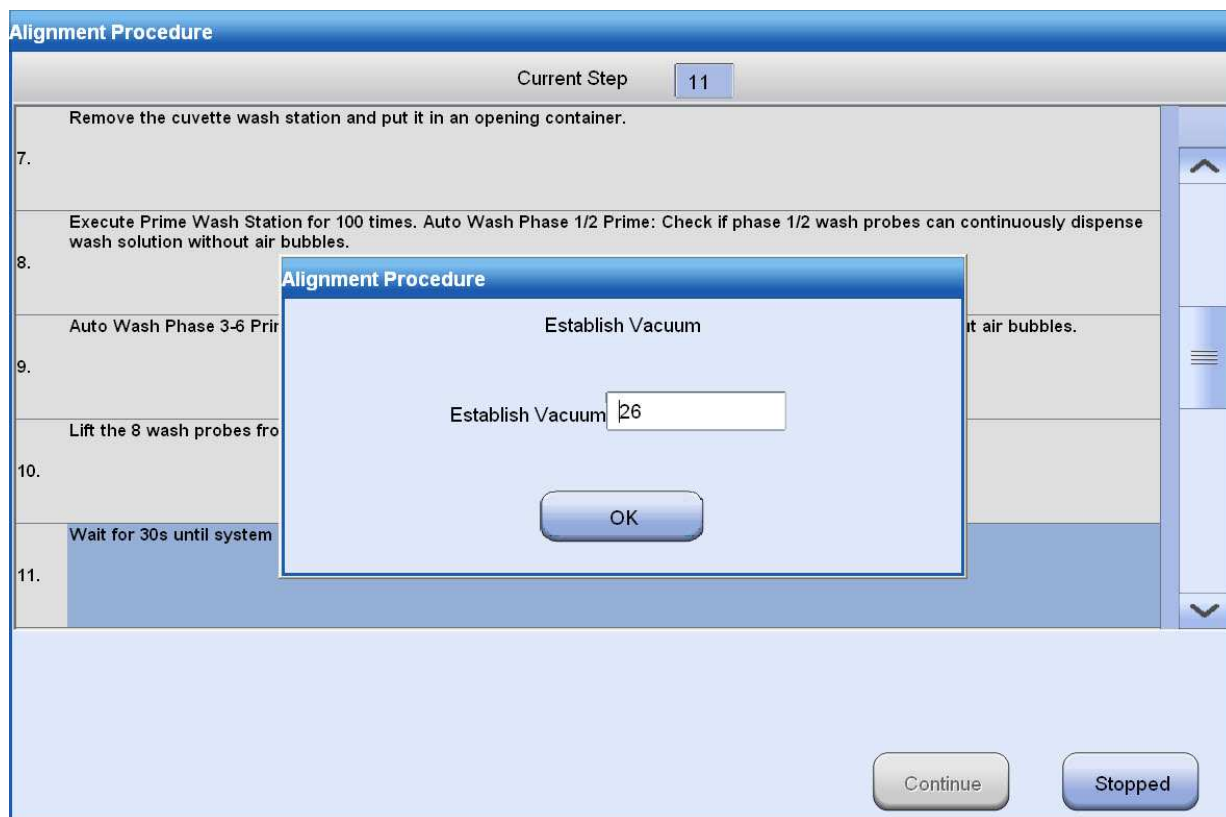
- 11.** In step 10, follow the screen instructions to lift the wash probes from the container. Then click **OK** to exit the popped window. Select **Continue** to proceed.



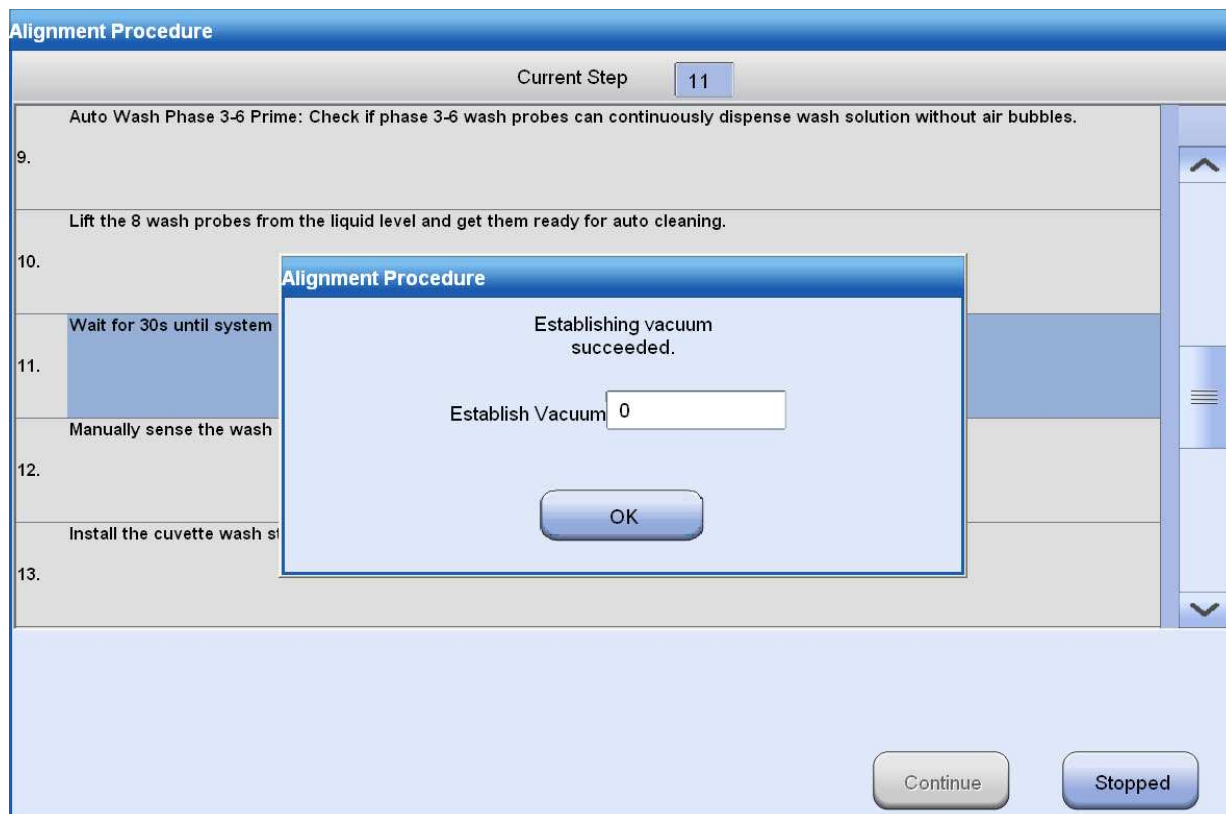
- 12.** In step 11, select **Continue** to establish vacuum.

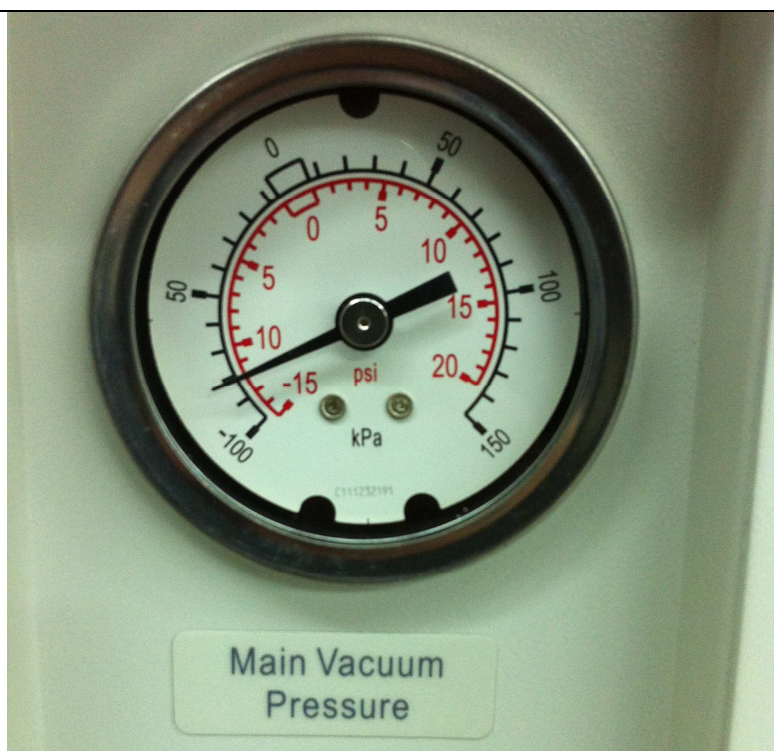


Wait unit countdown finishes. DO NOT click **OK** before time is up, otherwise you will return to step 10.



After it indicates successful, DO NOT click **OK**, please first check the readings of the vacuum gauge. The normal range is $\leq -70\text{KPa}$. Then click **OK** to proceed.





- 13.** In step 12, use your finger to feel the vacuum force of all eight probes. Then select **Continue** to proceed.

Alignment Procedure

Current Step 12

9.	Auto Wash Phase 3-6 Prime: Check if phase 3-6 wash probes can continuously dispense wash solution without air bubbles.
10.	Lift the 8 wash probes from the liquid level and get them ready for auto cleaning.
11.	Wait for 30s until system vacuum is established, and then check if the vacuum pressure is normal.
12.	Manually sense the wash probe force.
13.	Install the cuvette wash station on the drive.

Continue
Stopped



- 14.** In step 13, first remove the reaction disk cover then reinstall the cuvette wash station then click **Continue** to proceed.

Alignment Procedure

Current Step 13

9.	Auto Wash Phase 3-6 Prime: Check if phase 3-6 wash probes can continuously dispense wash solution without air bubbles.
10.	Lift the 8 wash probes from the liquid level and get them ready for auto cleaning.
11.	Wait for 30s until system vacuum is established, and then check if the vacuum pressure is normal.
12.	Manually sense the wash probe force.
13.	Install the cuvette wash station on the drive.

Continue
Stopped

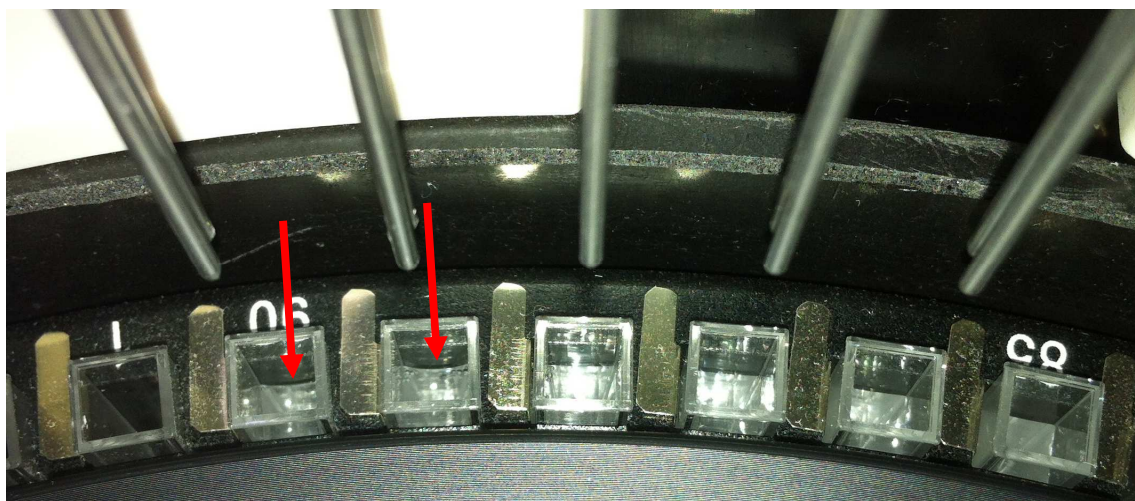
- 15.** In step 14, click **Continue** then wash station is performing cuvette wash for 10 times. Please visually check if there is liquid overflowing and if the fluid level in cuvettes is 5-10mm from the cuvette opening.

Alignment Procedure

Current Step 14

10.	Lift the 8 wash probes from the liquid level and get them ready for auto cleaning.
11.	Wait for 30s until system vacuum is established, and then check if the vacuum pressure is normal.
12.	Manually sense the wash probe force.
13.	Install the cuvette wash station on the drive.
14.	Cuvette normal wash is executed for 10 times. Check if cleaning fluid overflows the cuvettes, and if sufficient fluid is dispensed into cuvettes, about 5mm-10mm from cuvette opening.

Continue
Stopped



Check for overflow and make sure the liquid surfaces in these cuvettes are approximately in same height

16. In step 15, click **Continue** to reset the whole system for 3 times.

Alignment Procedure

Current Step: 15

13. Install the cuvette wash station on the drive.

14. Cuvette normal wash is executed for 10 times. Check if cleaning fluid overflows the cuvettes, and if sufficient fluid is dispensed into cuvettes, about 5mm-10mm from cuvette opening.

15. Priming is finished. The system resets for 3 times. The following actions are performed correctly: 1.

16. Fluidic prime is complete.

Continue Stopped

17. Step 16 indicates the priming is finished. Click **Continue** to exit the process.

Alignment Procedure

Current Step: 16

13. Install the cuvette wash station on the drive.

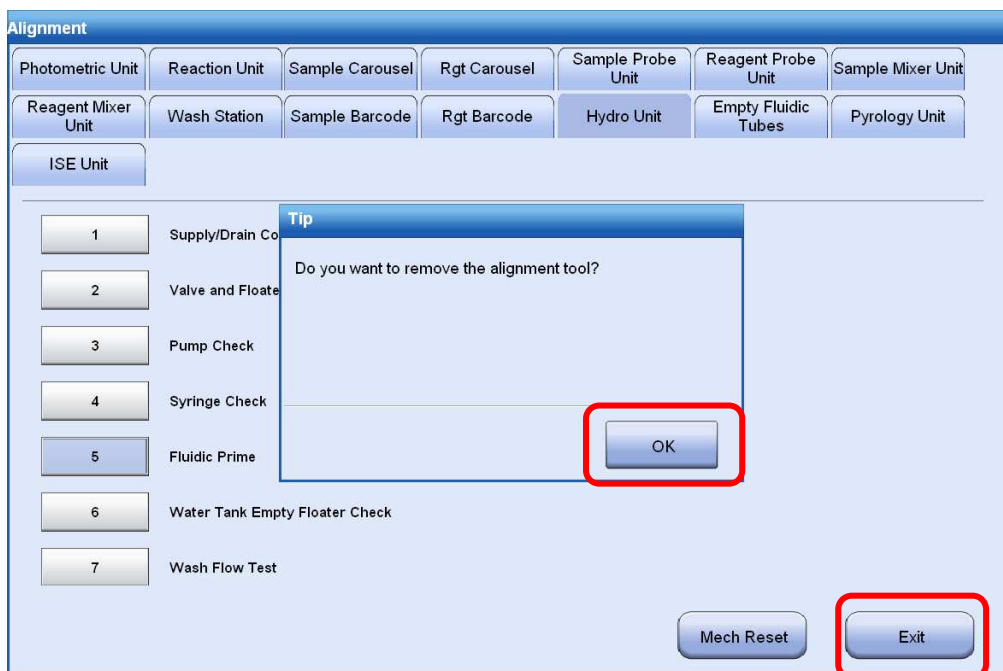
14. Cuvette normal wash is executed for 10 times. Check if cleaning fluid overflows the cuvettes, and if sufficient fluid is dispensed into cuvettes, about 5mm-10mm from cuvette opening.

15. Priming is finished. The system resets for 3 times. The following actions are performed correctly: 1.

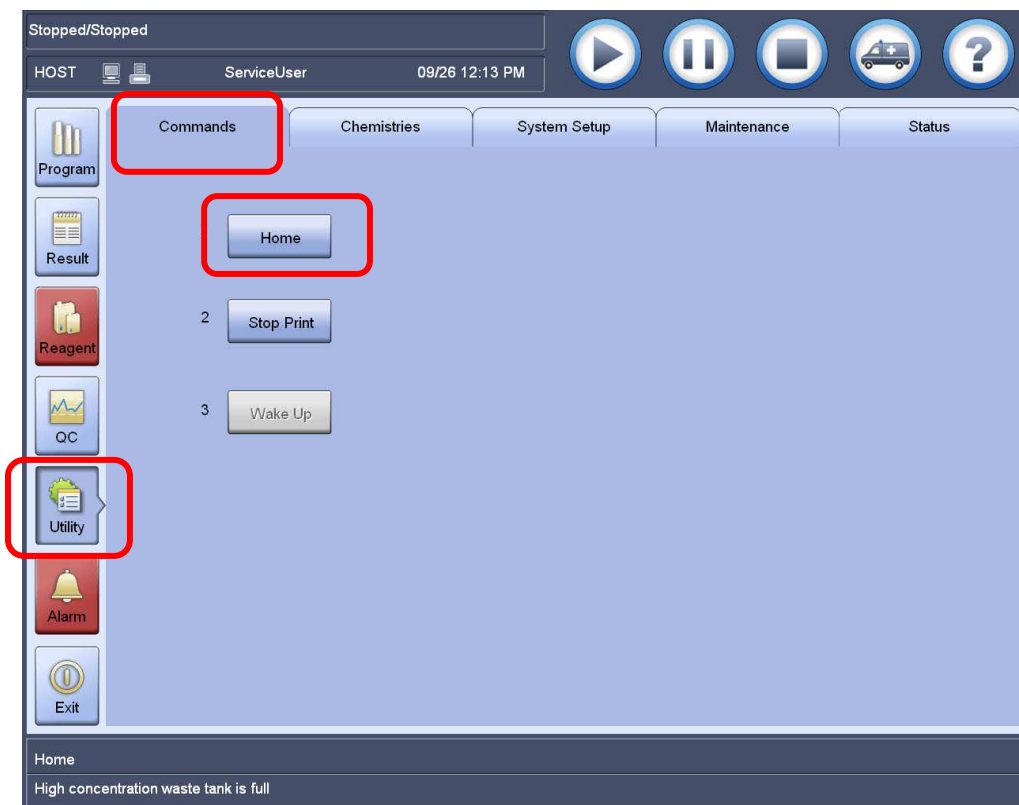
16. Fluidic prime is complete.

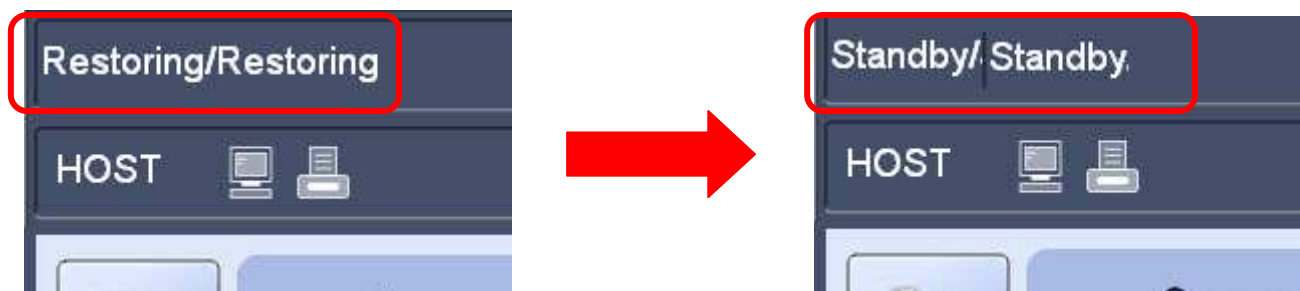
Continue Stopped

- 18.** Click **Exit** to return to the main menu. Then a window pops out to remind removing all alignment tools. Since we do not use any alignment tools, just click **OK**.



- 19.** After alignment, system status becomes “Stop”. You need to perform “Home” action to reset the system. Select **Utility** -> **Commands** -> **Home**. After Home, system status becomes standby and is ready for testing.

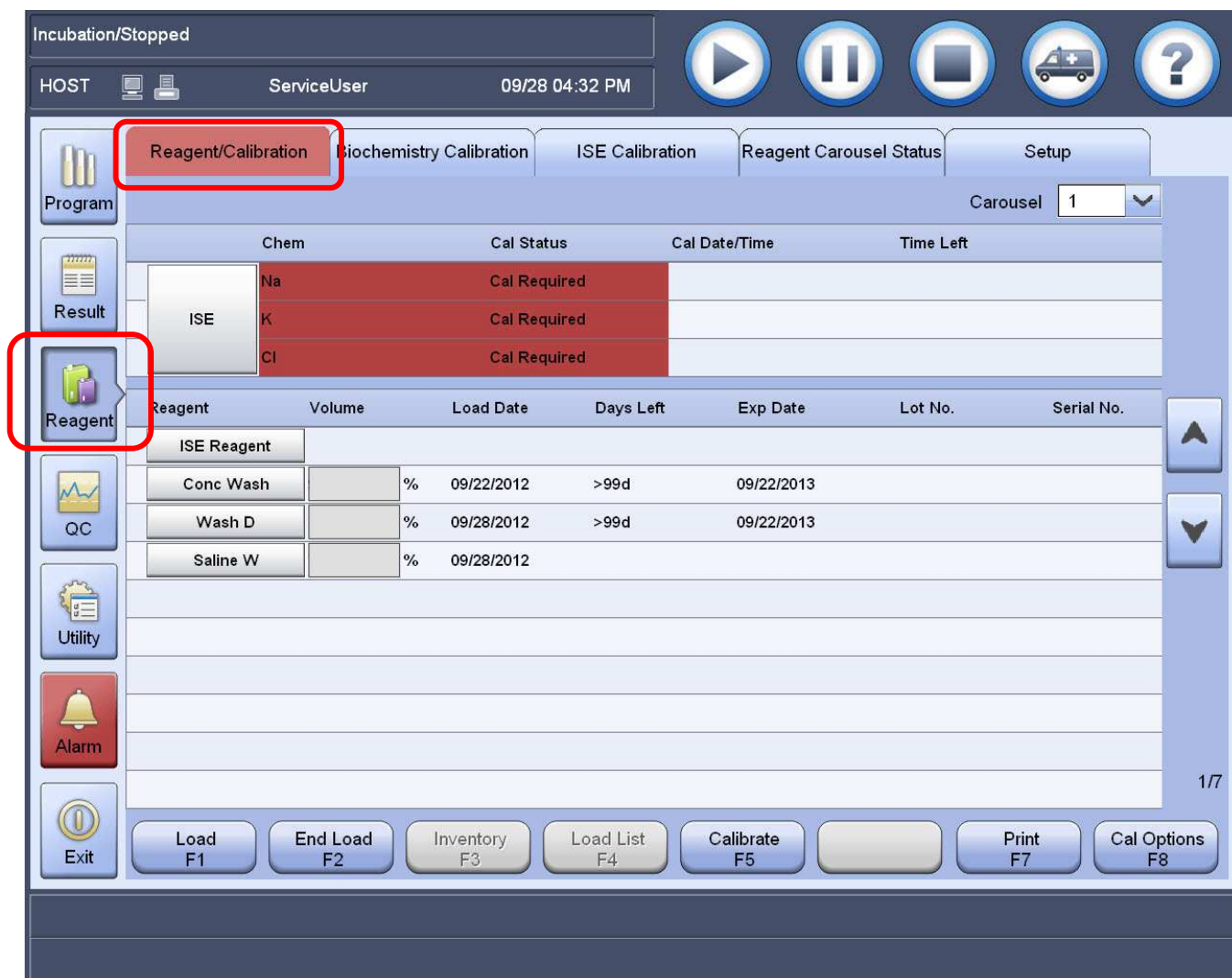




- 20.** Double check the reagent/sample probe, if there is no leakage from the tube/probe connection, you can reinstall the probe cover now.

3.10 Load Detergent into Reagent/Sample Disk

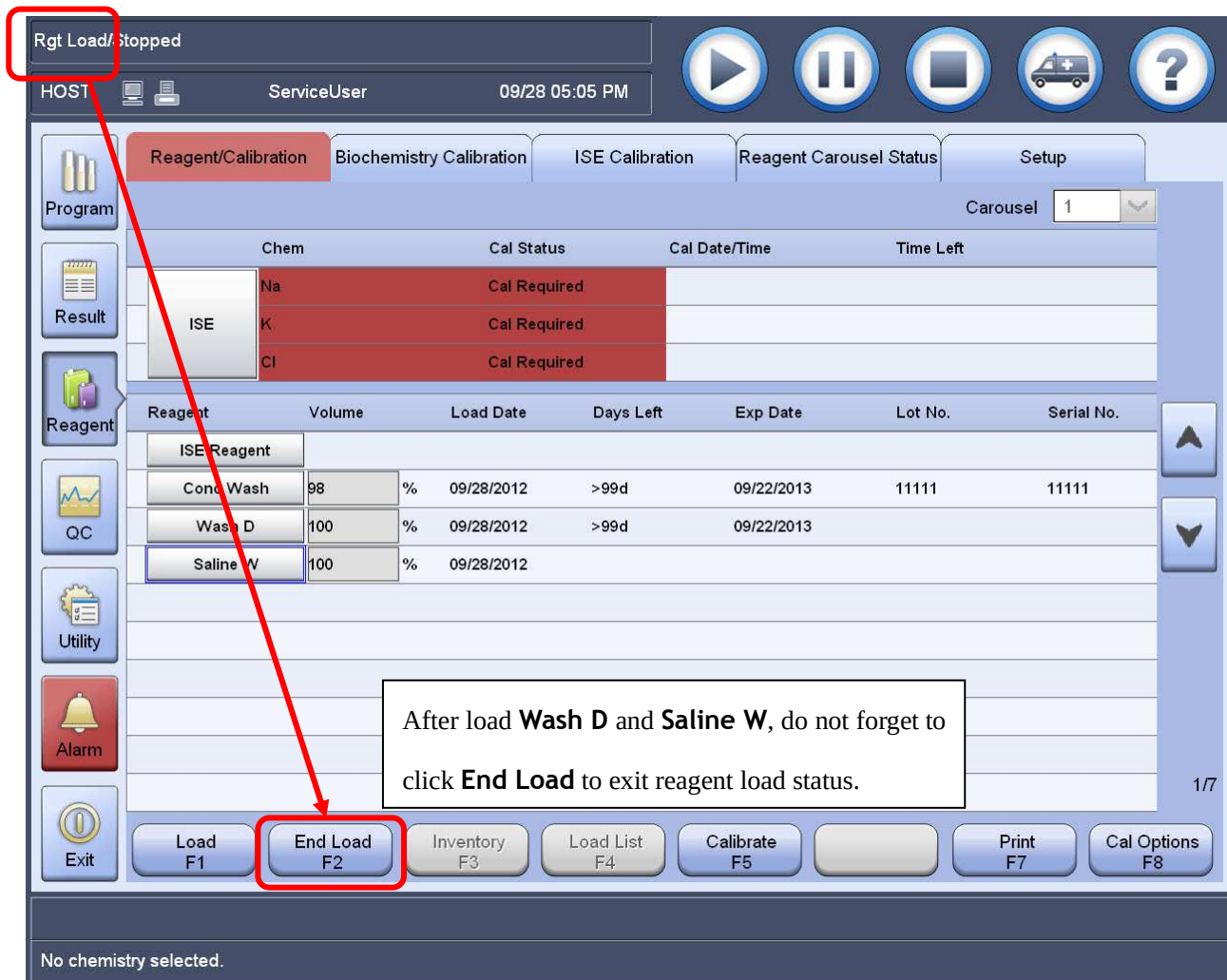
- 1.** Select **Reagent** -> **Reagent/Calibration**.



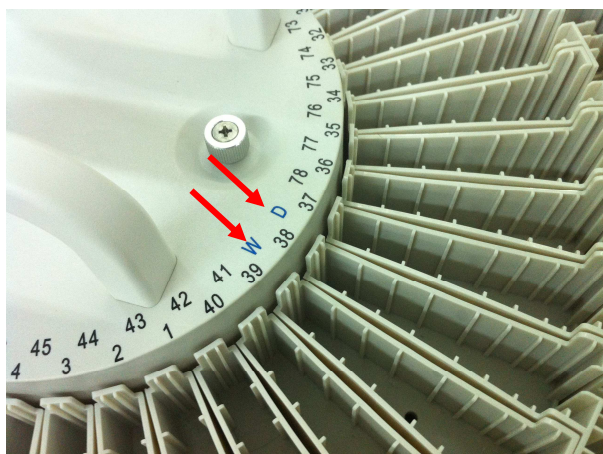
2. Click **Conc Wash** and then click **Load F1**. In popped window, finish all necessary information. Click **Load** to save the information and click **Exit** to return.

The screenshot displays the Mindray Biochemistry Global Technical Support software interface. The main window is titled 'Incubation/Stopped' and shows a 'HOST' status of 'ServiceUser' with a timestamp of '09/28 04:32 PM'. The interface includes a top navigation bar with icons for Play, Pause, Stop, and Help. Below this, there are tabs for 'Reagent/Calibration', 'Biochemistry Calibration', 'ISE Calibration', 'Reagent Carousel Status', and 'Setup'. The 'Reagent/Calibration' tab is active, showing a table with columns for 'Chem', 'Cal Status', 'Cal Date/Time', and 'Time Left'. The table lists 'Na', 'K', and 'Cl' with 'Cal Required' status. A 'Load Reagent' dialog box is open, showing fields for 'Chem' (Conc Wash), 'Volume' (100 %), 'Exp Date' (9/22/2013), 'Serial No.' (11111), and 'Lot No.' (11111). The 'Load' button is highlighted. The bottom of the interface features a row of function buttons: 'Exit', 'Load F1', 'End Load F2', 'Inventory F3', 'Load List F4', 'Calibrate F5', 'Print F7', and 'Cal Options F8'. The 'Load F1' button is highlighted. A status bar at the bottom indicates 'Enter reagent lot number(0-18 digits)'.

3. Repeat steps above to input information for **ISE reagent**, **Wash D** and **Saline**. Then click **End Load** to finish. **Note:** if you miss clicking **End Load**, system will always remain in “Rgt Load” status and you cannot perform any tests.



4. Fill a reagent bottle with diluted detergent CD80 (dilution ration 1:10). Put it in reagent disk **position**
D. Fill another reagent bottle with saline and put it in **position W**.



4 Acceptance Functional Test

4.1 Acceptance for Main Unit

1. Select **Utility** -> **Maintenance** -> **Maintenance**. Note: Main unit status should be “Standby”.

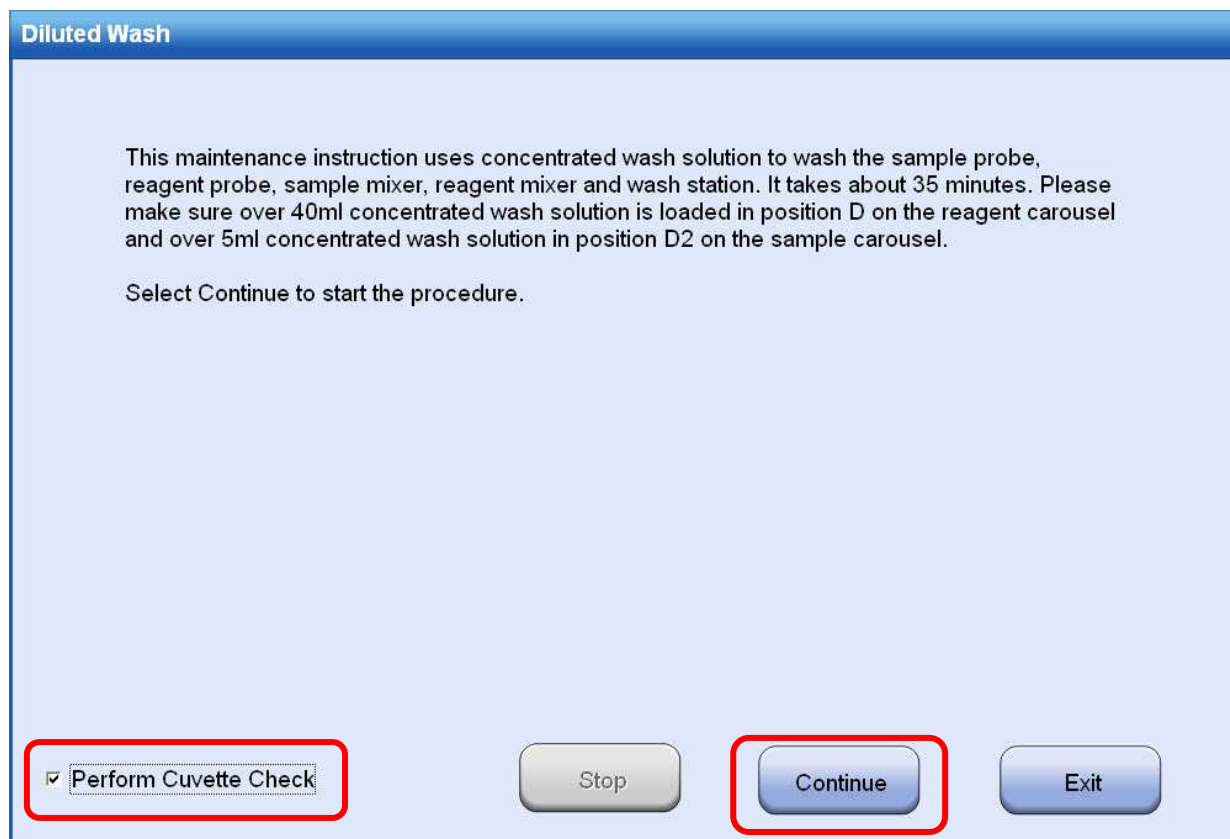
Otherwise you could not perform maintenance action. Do “Home” if necessary.



2. Select **Biochemistry Maintenance** -> **Diluted Wash**



3. Manually put more than 40ml CD80 in position D of reagent disk and more than 5ml CD80 in position D2 of sample disk. Then select **Perform Cuvette Check** and click **Continue** to proceed.



4. After the process is finished, cuvette check result is displayed. If you find any cuvette is marked red, please take it out and visually check for stains or cracks. Replace it if necessary. Click **Exit** to return.

Cuvette Check

Start

Last Check: YY-MM-DD

Stop

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70

To ensure photometer is in optimum status, run cuvette check at least once a week and replace or wash the ones marked yellow.

Result

Save

Exit

5. Click **Done** to exit. If the whole procedure is finished without any alarm message, it proves the hardware is OK.

Diluted Wash

The maintenance is complete.

Select Done to exit.

06:38

☒ Perform Cuvette Check

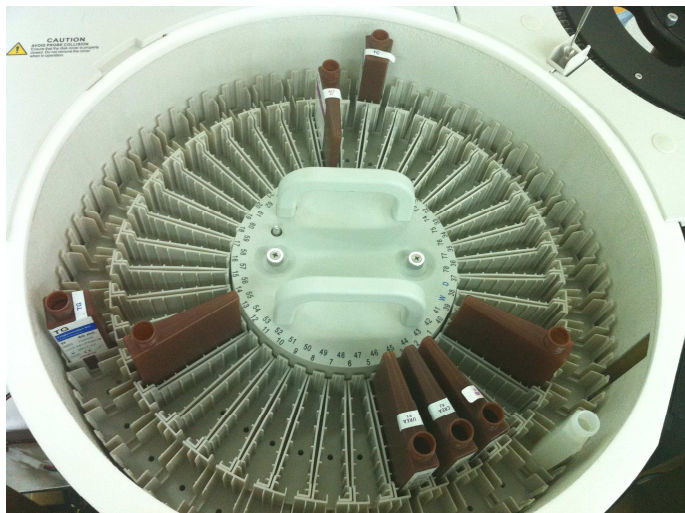
Stop

Done

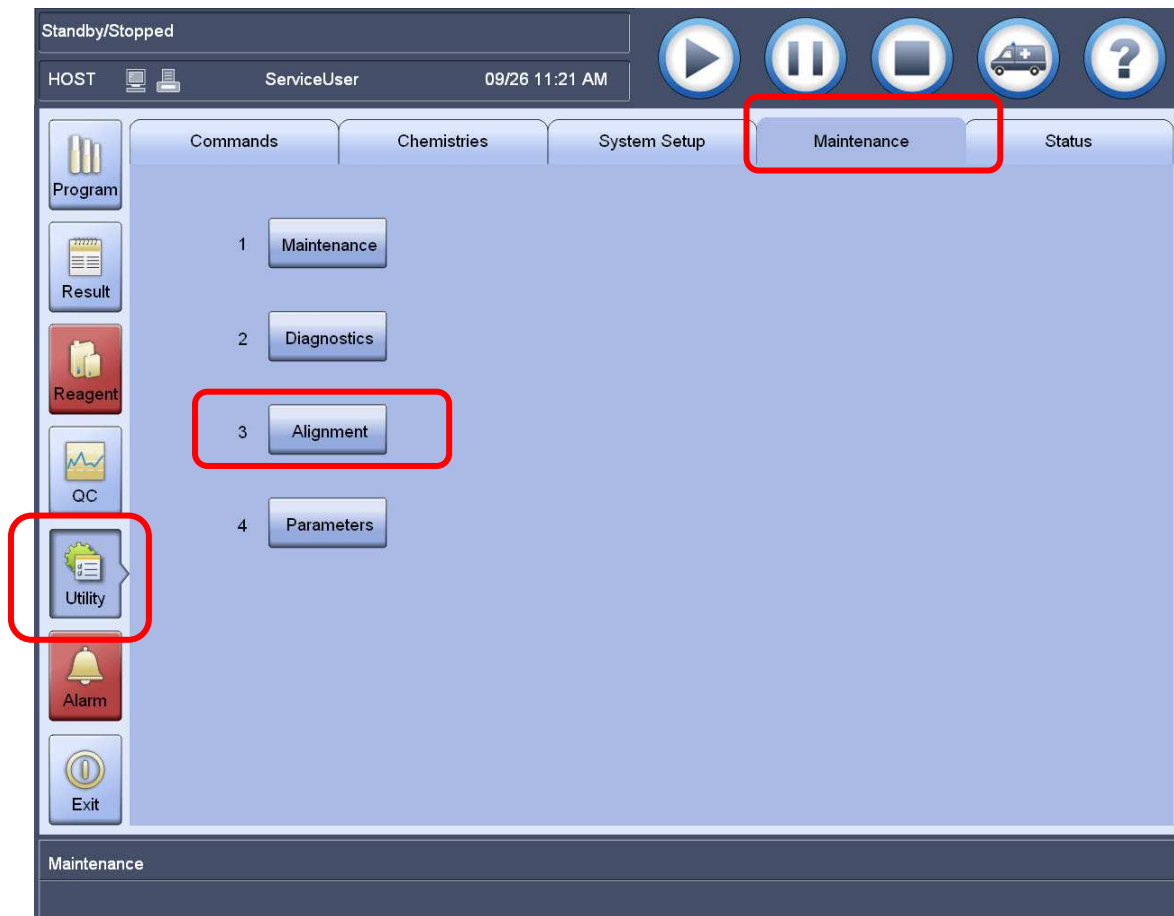
Exit

4.2 Acceptance for Reagent Barcode Reader (Optional)

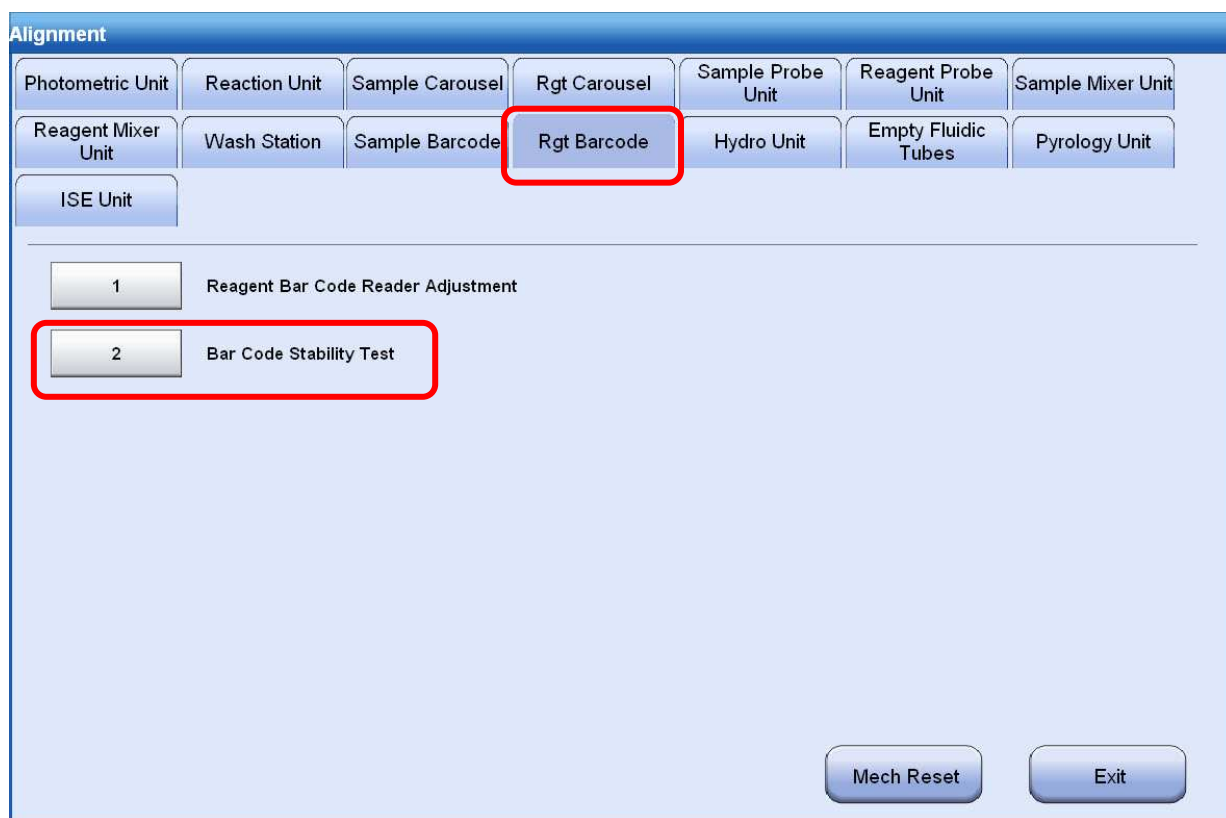
1. Load Mindray reagent in random positions of inner/outer cycle.



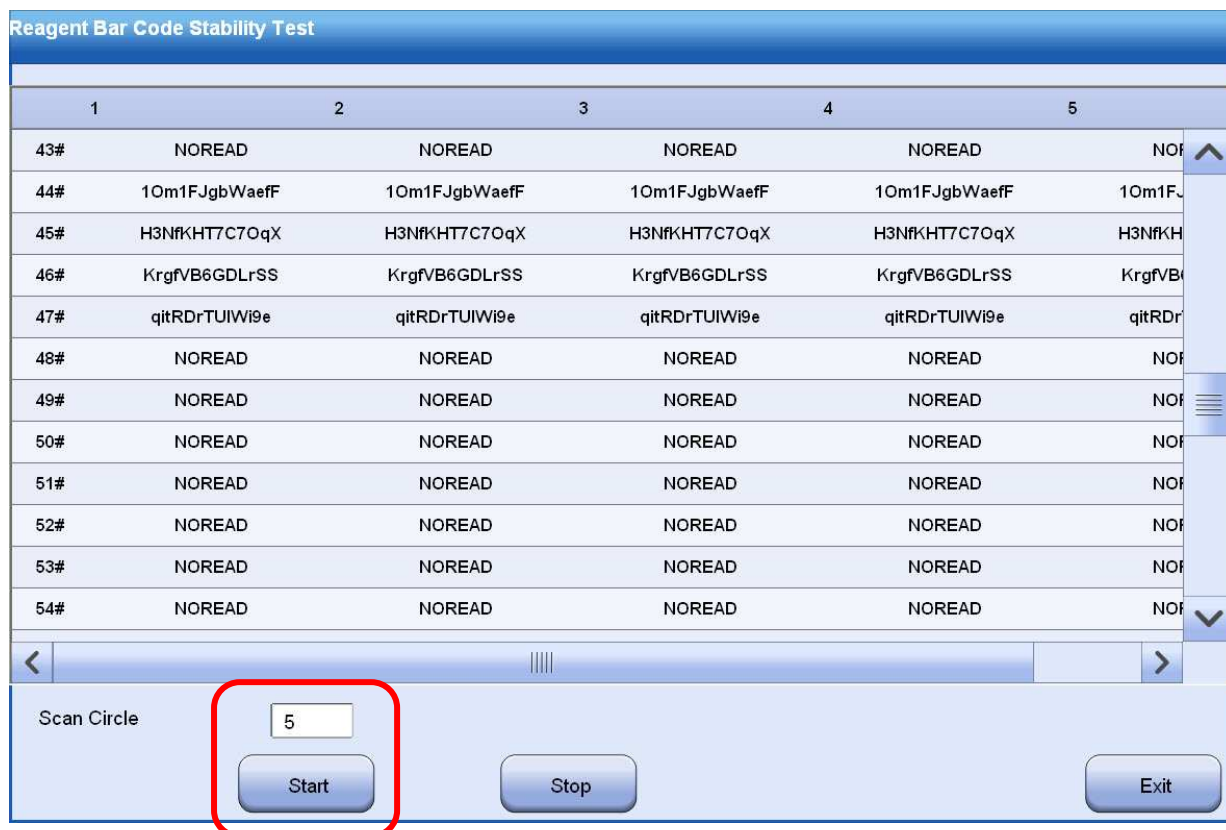
2. Select **Utility** -> **Maintenance** -> **Alignment**



3. Select Rgt Barcode -> Barcode Stability Test

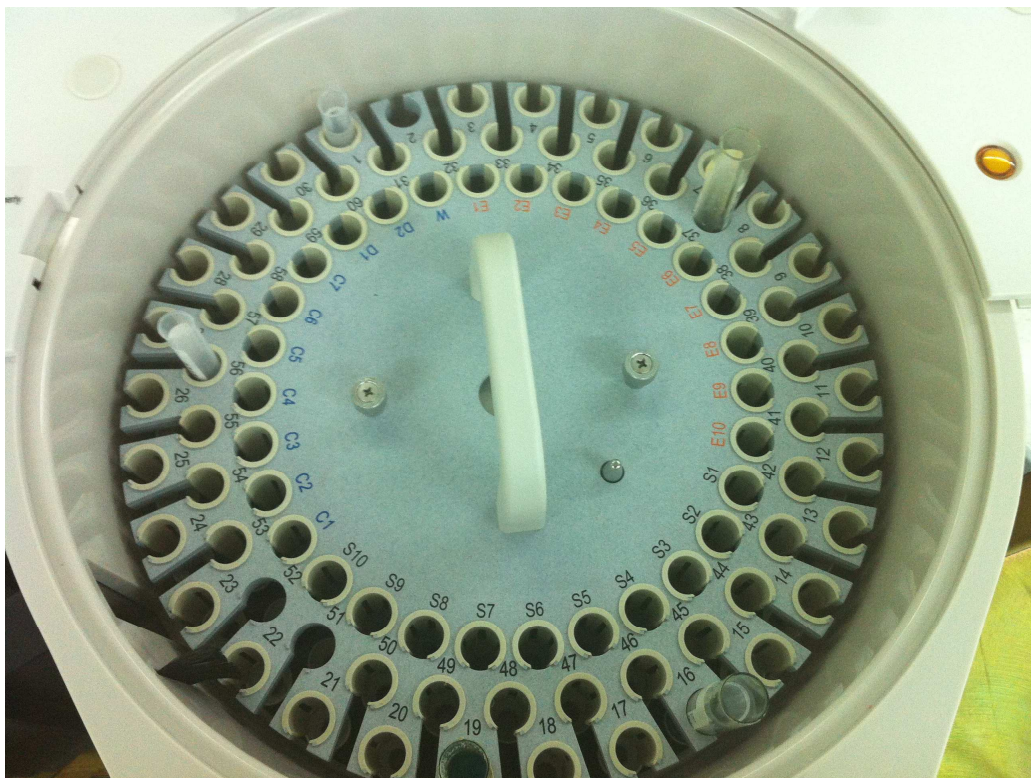


4. Input "5" in Scan Circle then click **Start**. Check if all reagents have readings. If not, you may need to re-adjust the scanning position.

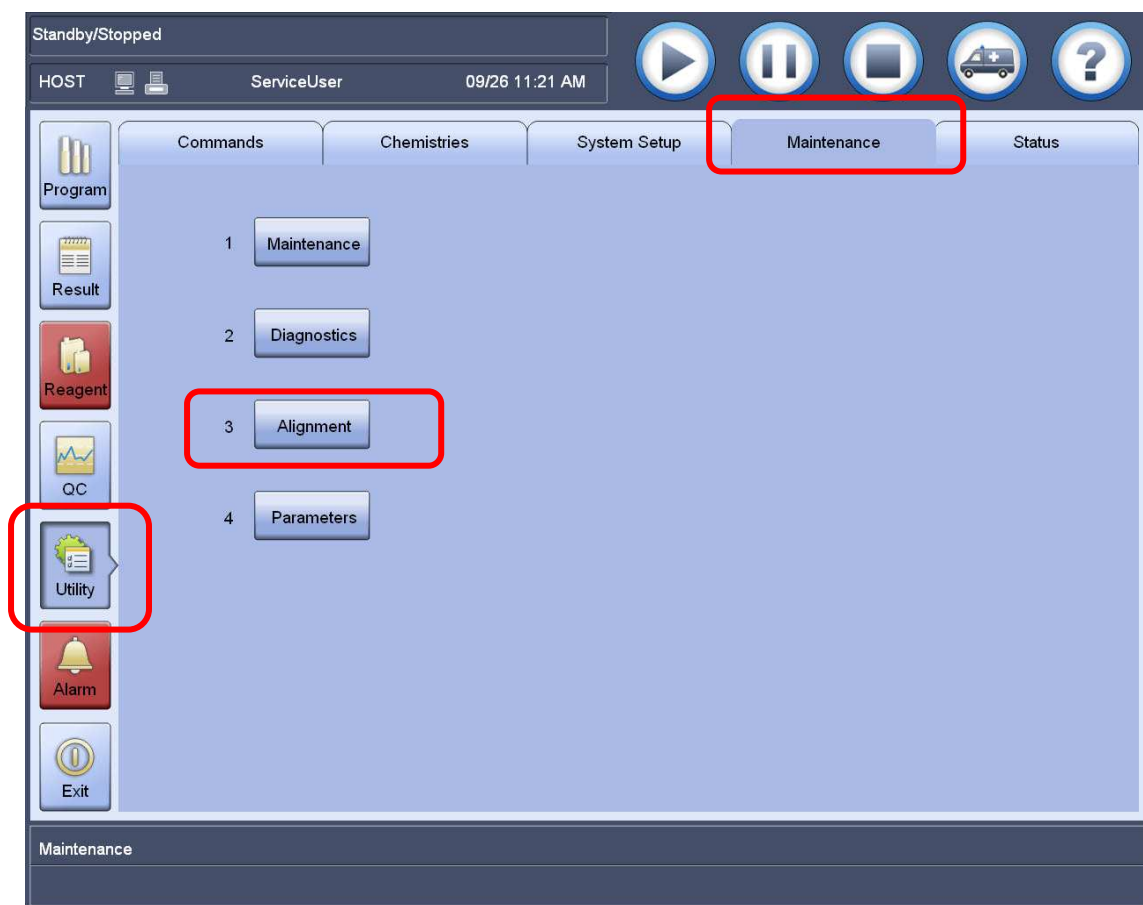


4.3 Acceptance for Sample Barcode Reader (Optional)

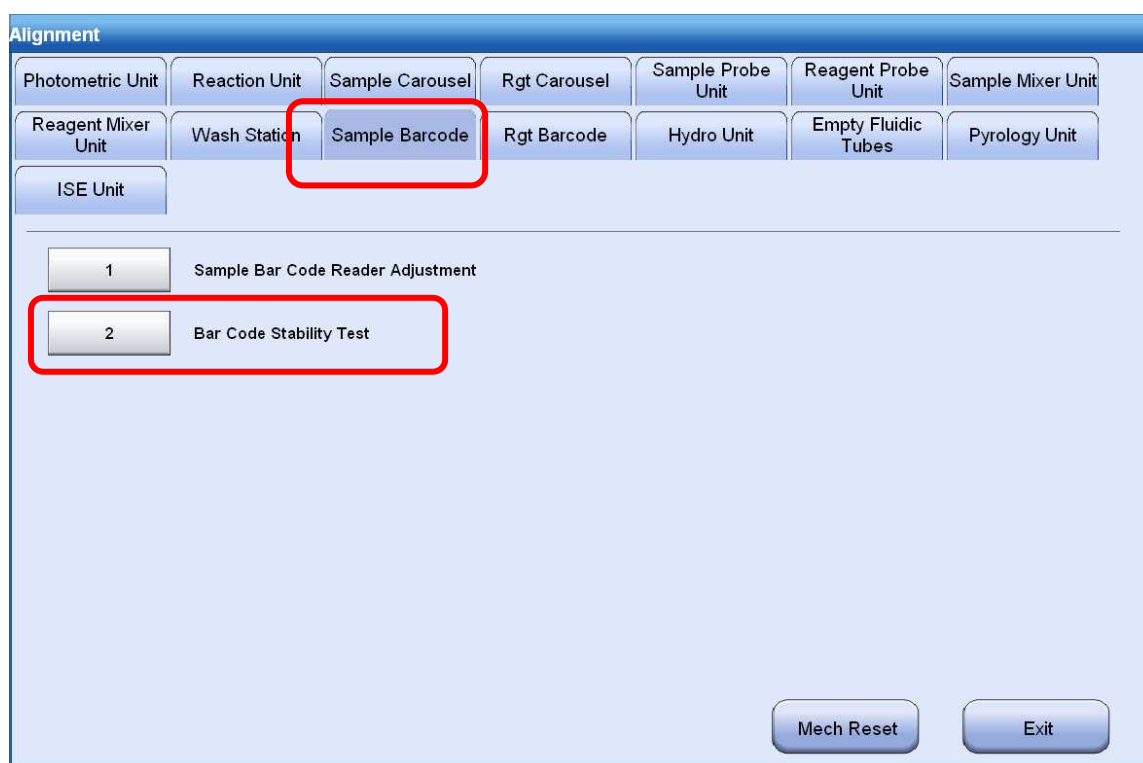
1. DO NOT use customer's barcode labels for testing since in alignment software ONLY supports code 128 type.
2. Detach labels from Mindray accessories, such as probes, acupuncture needle, and filter. Then attach these labels in sample tubes. Put these tubes in random positions of outer/middle cycle.



3. Select **Utility** -> **Maintenance** -> **Alignment**



4. Select **Sample Barcode** -> **Barcode Stability Test**



5. Input "5" in Scan Circle then click **Start**. Check if all these positions have readings. If not, you may need to re-adjust scanning positions.

Sample Bar Code Stability Test					
	1	2	3	4	5
1#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
2#	1234568111	1234568111	1234568111	1234568111	1234568111
3#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
4#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
5#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
6#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
7#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
8#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
9#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
10#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
11#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD
12#	NOREAD	NOREAD	NOREAD	NOREAD	NOREAD

Scan Circle

Start

Stop

Exit

4.4 Re-arrangement

After the performance check, the equipment is ready for testing. Please Re-arrange the external tubings and tanks, relocate the machine if necessary, then rotate the four anchors down to the floor so it is in a secured status.



5 Appendix

5.1 Correspondence of Altitude - Atmospheric Pressure

Altitude(m)	Atmospheric Pressure(kPa)	Atmospheric Pressure(psi)
-400	105.9	15.4
-300	104.8	15.2
-200	103.6	15.0
-100	102.5	14.9
0	101.3	14.7
100	100.2	14.5
200	99.1	14.4
300	98.0	14.2
400	96.9	14.0
500	95.8	13.9
600	94.7	13.7
700	93.6	13.6
800	92.6	13.4
900	91.5	13.3
1000	90.5	13.1
1100	89.4	13.0
1200	88.4	12.8
1300	87.4	12.7
1400	86.4	12.5
1500	85.4	12.4
1600	84.4	12.2
1700	83.4	12.1

Altitude(m)	Atmospheric Pressure(kPa)	Atmospheric Pressure(psi)
1800	82.4	12.0
1900	81.5	11.8
2000	80.5	11.7
2100	79.6	11.5
2200	78.6	11.4
2300	77.7	11.3
2400	76.8	11.1
2500	75.9	11.0
2600	75.0	10.9
2700	74.1	10.7
2800	73.2	10.6
2900	72.3	10.5
3000	71.4	10.4
3100	70.6	10.2
3200	69.7	10.1
3300	68.9	10.0
3400	68.0	9.9
3500	67.2	9.7
3600	66.4	9.6
3700	65.6	9.5
3800	64.8	9.4
3900	64.0	9.3
4000	63.2	9.2
4100	62.4	9.0
4200	61.6	8.9
4300	60.8	8.8
4400	60.1	8.7
4500	59.3	8.6
4600	58.6	8.5
4700	57.8	8.4
4800	57.1	8.3
4900	56.4	8.2
5000	55.6	8.1